

X-Ray Calc 3

User Manual

Version 11

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Contents

1. Introduction
2. Quick Start
3. Installation & First Launch
4. Application Overview
5. Quick Reference
6. Projects
7. Defining a Model
8. Experimental Data
9. Calculation
10. Fitting (Optimization)
11. Results & Export
12. Charts & Visualization
13. Profile Extensions
14. Tools
15. Application Settings
16. Universal Mirror Optimizer v3.7
17. MCP Server (XRC_MCP)
18. File Formats
19. Illustrative Examples
20. Troubleshooting
21. Source Code & Git
22. Online Wiki

1 Introduction

What X-Ray Calc 3 does, the calculation method, and the reference publication

X-Ray Calc 3 (XRC3) is an open-source software package for simulating X-ray reflectivity (XRR) and solving the inverse problem of reconstructing film structures from measured XRR curves. It is designed for researchers working with X-ray optics, synchrotron radiation, and thin-film coatings.

The software employs Parratt's exact recursive method based on the Fresnel equations to calculate XRR and incorporates specialized tools for modeling periodic multilayer structures (PMMs). Refractive indices of materials are computed using the Henke tables of atomic scattering factors (Henke *et al.*, 1993). The computational algorithm has been validated against the *IMD* software (Windt, 1998), with accumulated variances consistently below 0.5%.

The application allows you to:

- Build multilayer structural models composed of stacks and individual layers
- Calculate theoretical X-ray reflectivity curves as a function of incidence angle (θ) or wavelength (λ)
- Load experimental reflectivity data and compare it with theoretical predictions
- Automatically fit model parameters to experimental data using the Shake LFPSO (Lévy Flight Particle Swarm Optimization) algorithm
- Visualize results through interactive charts with logarithmic/linear scaling
- Define parameter profiles (polynomial or tabulated) across periodic structures
- Manage materials via the built-in Henke atomic form factor database

Reference: Penkov, O. V., Li, M., Mikki, S., Devizenko, A. & Kopylets, I. (2024). *X-Ray Calc 3: improved software for simulation and inverse problem solving for X-ray reflectivity*. J. Appl. Cryst. **57**, 555–566.
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See Also

- [Quick Start](#)
- [Fitting \(Optimization\)](#)
- [Source Code & Git](#)

2 Quick Start

A hands-on tour: open a demo project, build a model from scratch, calculate and export

This section provides a hands-on introduction to X-Ray Calc 3. The distribution includes several demonstration projects in the **Examples** folder — open one to explore the interface before building your own models.

Exploring the Demo Projects

1. Click  **Open** and navigate to the **Examples** folder.
2. Select a project file (`.xrcx`).
3. Go to the **Computation** tab and click **Run** (or press **F5**). You can also use **Calc All (F12)** to calculate all models at once.

Navigating the Interface

- **Double-click a model name** in the Project Items list to open its Properties dialog (name, color, description). The same dialog is available from the right-click context menu.
- **Double-click a layer** to open the Layer Properties dialog, where you can change material, thickness, roughness, and density.
- **Double-click a stack header** to open the Stack Properties dialog, where you can rename the stack and change the number of periods.
- Use the **Stack** and **Layer** toolbar panels to add, insert, or delete structural elements.

Creating a Model from Scratch

1. Create a new project (**File > New Project**).
2. Select **Model 1** in the project tree.
3. Add a stack: **Structure > Add Stack** (or use the toolbar). In the dialog, set the stack name (e.g., “Main”) and the number of periods (default: 1 for non-periodic structures).
4. Select the stack by clicking its title in the structure panel.
5. Add layers: click  **Add Layer** (or **Structure > Add Layer**). Fill in the material (click “...” to select from the Materials Database), thickness, roughness, and density. Repeat for each layer.
6. Calculate the reflectivity: press **F5** (or **Calc > Calculate**).

Calculation Settings

- **Mode:** Reflectivity can be calculated as a function of grazing angle or wavelength.
- **Polarization:** **sp** is the unpolarized average $(R_s + R_p)/2$ and takes about twice as long as **s**. At small grazing angles the two differ little (**sp** is lower than **s** by about 1.5 % at 5° and 6 % at 10°), so **s** is the usual choice for hard X-ray XRR; use **sp** for an unpolarized source at larger grazing angles.
- **Number of points (N):** Controls calculation precision. Calculation time scales linearly with N.

- **Angle range:** Set with θ_1 and θ_2 . Check **2 θ** for θ -2 θ geometry. **d θ** is the angular resolution: the FWHM (degrees, in θ) of the Gaussian the calculated curve is convolved with (0 = none).
 - **Scale:** Toggle between **Linear** and **Log** scale using the toolbar button. Set a **background level** if needed.
 - **Legend:** Click items in the chart legend to toggle curve visibility.
-

Exporting Results

- The **Export** button on the Result panel saves the calculated curve to an ASCII file or copies it to the clipboard.
- The **Save** button on the Plot panel saves the chart as a bitmap image.

See Also

- [Defining a Model](#)
- [Calculation](#)
- [Illustrative Examples](#)

3 Installation & First Launch

System requirements, the working directory, portable mode, and file association

System Requirements

- Windows 10 or later (64-bit recommended)
- Minimum 4 GB RAM (8 GB recommended for large models)
- Multi-core CPU recommended for parallel computation

Data Storage

On first launch, X-Ray Calc 3 creates a working directory:

- **Default location:** %APPDATA%\X-RayCalc3\
- **Portable mode:** If a file named uselocaldata exists in the application folder, all data is stored locally next to the executable.

The working directory contains:

FILE / FOLDER	PURPOSE
xrc3.ini	Application settings
Henke/	Atomic form factor tables
Backup/	Auto-save and backup projects
Output/	Fitting results (when auto-save is enabled)
Benchmark/	Benchmark model files
Jobs/	Batch job definitions

File Association

To associate .xrcx project files with X-Ray Calc 3, go to **File > Settings > Paths and Files** and click **Register file associations (xrcx)**.

See Also

- [Application Settings > Paths and Files](#)
- [File Formats](#)
- [Troubleshooting](#)

4 Application Overview

Main window layout, the three columns, status bar, and the menu bar

Main Window Layout

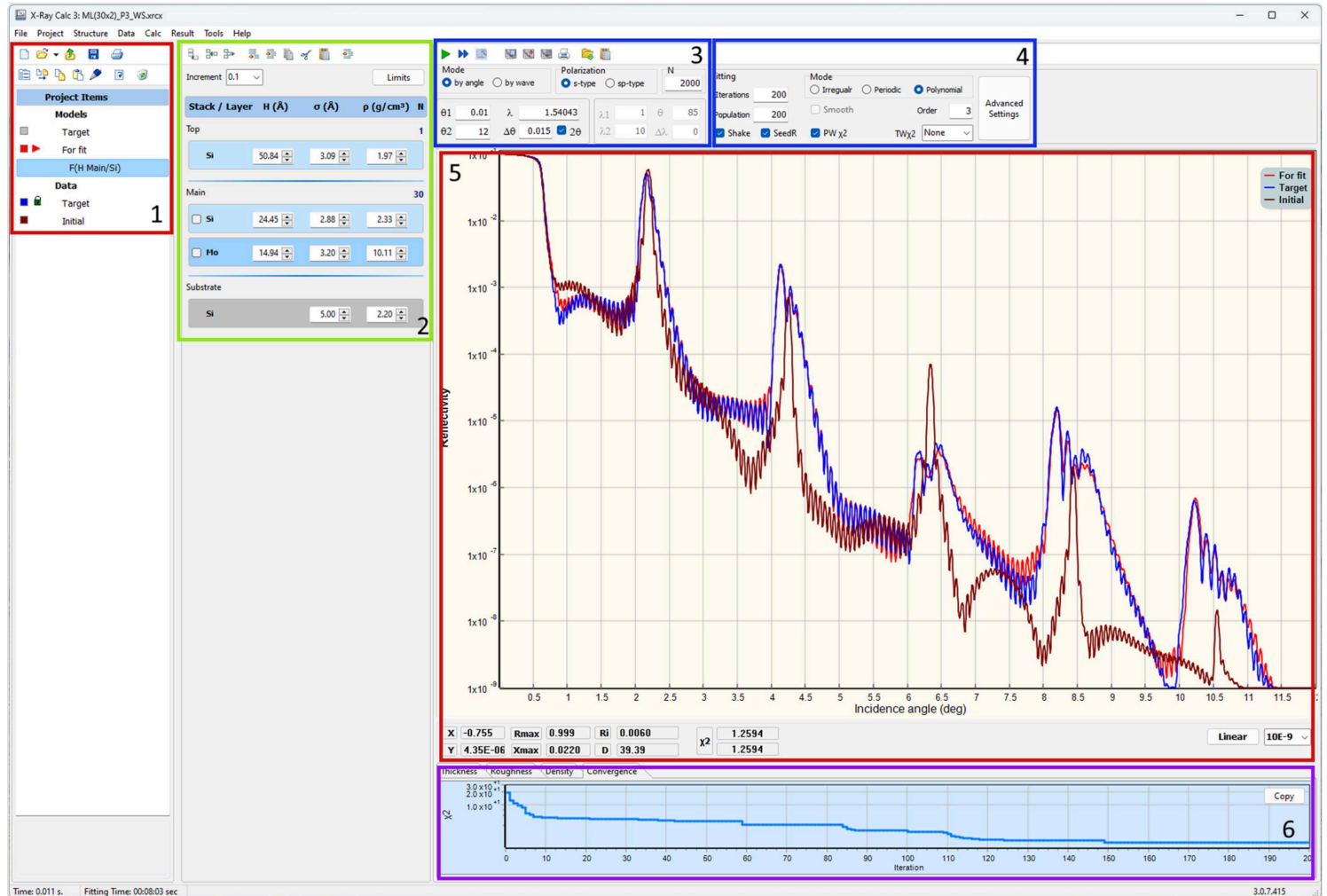


Figure 1. Screenshot of the XRC3 GUI. (1) Project tree. (2) Structure panel. (3) XRR calculation parameters. (4) Fitting parameters. (5) XRR curves. (6) Distributions / Convergence panel. (From Penkov et al., 2024.)

The primary window comprises four key panels (see Figure 1). The first panel, **Project Items** (1), showcases the contents of the current project. A project can organize an unlimited number of theoretical models and experimental curves into well-structured folders. The **Structure panel** (2) functions as a visual depiction of the layered model and provides easy access to all parameters linked with the structural model. The right side contains **XRR calculation parameters** (3) and **Fitting parameters** (4) at the top, the main **XRR curve plot** (5) in the center, and **Distributions / Convergence charts** (6) at the bottom.

Left Column — Project Panel

- **File Toolbar:** New, Open, Reopen, Save, Print
- **Project Toolbar:** Add Model, Export, Copy, Paste, Edit Properties, Add Extension, Copy Image, Print, Delete

- **Project Tree:** Hierarchical view of all models, data sets, and folders in the current project
- **Description:** Read-only text memo showing the description of the selected item

Middle Column — Structure Panel

- **Structure Toolbar:** Stack operations ( Add,  Insert,  Delete) and Layer operations ( Add,  Insert,  Copy,  Cut,  Paste,  Delete)
- **Increment Selector:** Step size for quick parameter adjustment (10, 5, 1, 0.25, 0.1, 0.01)
- **Set Fit Limits Button:** Opens the parameter bounds editor for fitting
- **Visual Structure Editor:** Interactive graphical representation of the layer structure — shows Ambient, Stacks (with N repetitions), individual Layers (with H, σ , ρ values), and Substrate. Double-click elements to edit properties; click parameter values for in-place editing.

Right Column — Working Area

- **Calculation Toolbar:**  Load Data,  Paste Data,  Calculate,  Save Result,  Copy Result,  Batch Execute,  Auto Fit, Stop
- **Settings Panel:** Calculation mode and parameters (left half) alongside Fitting mode and parameters (right half)
- **Main Chart:** Interactive reflectivity plot (θ or λ vs. reflectivity)
- **Chart Info Bar:** Cursor position, peak detection, area integration, χ^2 values and the scale they are taken at
- **Chart Pages:** Tabbed sub-charts for parameter distributions (Thickness, Roughness, Density, Profile, Fitting Progress)

Status Bar

Displays calculation time, fitting time, application version, and platform (x64).

Menu Bar

MENU	PURPOSE
File	Project management (New, Open, Save, Settings, Exit)
Project	Model operations (New Model, Duplicate, Copy, Paste, Add Folder)
Structure	Stack and layer editing (Add, Insert, Delete, Copy, Paste, Cut)
Data	Experimental data operations (Load, Paste, Normalize, Smooth, Trim, Assess XRR quality, Export)
Calc	Calculation and fitting (Calculate, Calculate All, Auto Fitting, Batch Jobs, Benchmark)
Result	Export results (Save, Copy to Clipboard, Copy as BMP/WMF)
Tools	Material management (Create Material, Edit Henke Table)
Help	User Manual, About dialog

See Also

- [Quick Reference](#)
- [Projects](#)
- [Charts & Visualization](#)

5 Quick Reference

Toolbar buttons, the structure toolbar, and keyboard shortcuts on one page

Toolbar Quick Reference

BUTTON	ACTION
 New	Create a new project
 Open	Open an existing project
 Save	Save the current project
 Print	Print the current chart
 Add Model	Add a new model to the project
 Load Data	Load experimental data from file
 Paste Data	Paste experimental data from clipboard
 Calc Run	Calculate the selected model
Stop	Abort the current calculation or fitting
 Fast Forward	Start auto-fitting
 Resume Fitting	Continue fitting from the last result, holding frozen parameters — see Staged Fitting
 Result Save	Save calculation results to file
 Copy Result	Copy results to clipboard
Scale	Toggle linear/logarithmic chart scale
 Delete	Remove the selected item

Structure Toolbar

BUTTON	ACTION
 Period +	Add a new stack
 Period Insert	Insert a stack at the current position
 Period -	Delete the selected stack
 Layer +	Add a new layer
 Layer Insert	Insert a layer at the current position
 Layer Copy	Copy the selected layer
 Layer Cut	Cut the selected layer
 Layer Paste	Paste the copied layer
 Layer -	Delete the selected layer
Set Fit Limits	Open the parameter bounds editor
Increment	Set the step size for parameter adjustments

Keyboard Shortcuts

KEY	ACTION
F1	Open User Manual

See Also

- [Application Overview](#)
- [Defining a Model](#)

6 Projects

Project files, creating, opening and saving, auto-save, locking, and the project tree

Project Files

Projects are saved as `.xrcx` files. Internally, an `.xrcx` file is a ZIP archive containing:

- `project.dsc` — project metadata
- `params.dsc` — model structure definitions
- Associated experimental data files

Creating a New Project

1. Choose **File > New Project** (or click the  **New** toolbar button).
2. An empty project is created with no models or data.

Opening a Project

- **File > Open Project** — browse for a `.xrcx` file
- **File > Recent Projects** — select from up to 10 recently opened projects
-  **Open toolbar button** — click the dropdown arrow for recent projects

Saving a Project

- **File > Save Project** — save to the current file
- **File > Save Project As** — save with a new name/location

Auto-Save

When enabled in Settings, the application periodically saves a backup to `AutoSave.xrcx` in the temp directory. This provides crash recovery — on next launch, you will be prompted to restore the auto-saved project.

Project Locking

When a project is open, a lock file (`temp.lock`) prevents simultaneous access from another instance. The lock is released when the project is closed.

Project Tree Organization

The project tree shows items in a hierarchy:

- **Model Group:** Contains all structural models

- Individual models (each with its own layer structure and calculation parameters)
- Extensions (profile functions attached to models)
- **Data Group:** Contains experimental data sets
- **Folders:** User-created organizational folders

Tip: Right-click the project tree for context menu operations.

See Also

- [File Formats > Project File](#)
- [Defining a Model](#)
- [Application Settings](#)

7 Defining a Model

The structure editor, stacks and layers, materials, quick editing, undo, and the JSON view

Creating a Model

- 1. Choose **Project > New Model** or click the  **Add Model** toolbar button.
- 2. A new model appears in the project tree with a default empty structure.

Duplicating a Model

Select a model in the project tree and choose **Project > Duplicate Model** to create an exact copy, including all layers, stacks, and parameters.

The Structure Editor

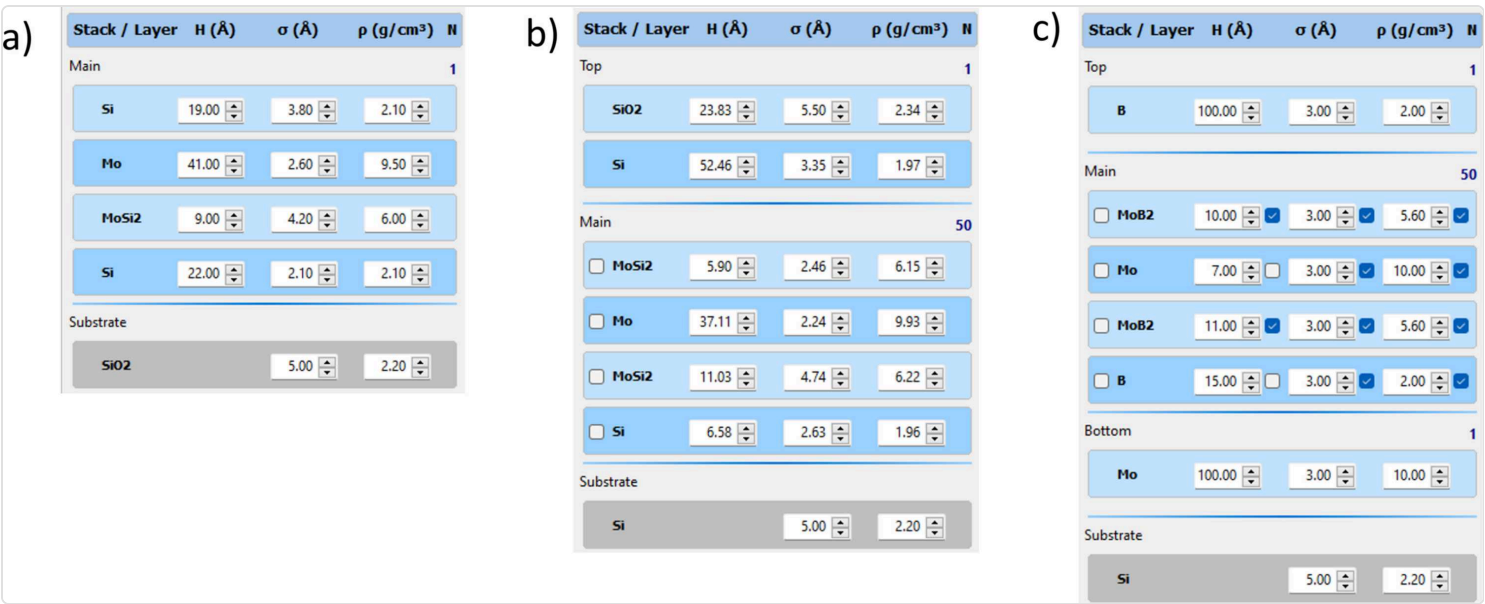


Figure 2. Examples of different structure types. (a) Regular structure. (b) Periodic structure. (c) Periodic structure with distributions. (From Penkov et al., 2024.)

The structure editor is a visual panel in the middle column that displays the current model's layer structure. It shows:

- **Stacks** — repeating periodic units containing one or more layers
- **Single layers** — stacks with $N = 1$; every layer belongs to a stack
- **Substrate** (bottom) — the base material

The medium above the structure is always vacuum ($\epsilon = 1$); it is not shown and cannot be edited.

Each element in the structure is displayed as a colored block with its parameters.

Terminology

The following terminology is used throughout XRC3 (as defined in Penkov *et al.*, 2024):

- **Material** — the chemical composition of a layer (e.g. Mo, SiO₂, B₄C). Used to look up atomic scattering factors, atomic weights and table densities in the embedded database. Cannot be changed during fitting.
- **Layer** — the basic geometrical entity. A layer consists of a material and has three parameters: thickness (H), roughness (σ) and density (ρ). All three are adjustable during fitting.
- **Stack** — a set of layers. It can be single (non-repeating) or periodic (repeated N times).
- **Structure** — a set of stacks, including a substrate. The substrate is a special stack consisting of a single layer with infinite thickness.

Stacks (Periods)

A stack represents a periodic multilayer structure — a group of layers repeated N times.

Adding a stack

1. Select the model in the project tree.
2. Choose **Structure > Add Stack** or use the structure toolbar.
3. The Stack Properties dialog opens with:
 - **Stack name:** A label for the stack
 - **N:** Number of repetitions (periods)

Editing a stack

Double-click the stack header in the structure editor to open the Stack Properties dialog.

Freezing a stack v3.8.2

Right-click a stack and choose **Freeze stack** to hold all three parameters (H, σ , ρ) of every layer in it during the next fit, or **Thaw stack** to release them. The stack header then reads *[frozen]*, or *[3/6 frozen]* when only part of it is held.

This is the same state the Limits dialog edits in its **Fix** columns, offered where you are already looking when you decide a stack is settled. See [Staged Fitting](#).

Layers

Each layer has three primary parameters:

PARAMETER	SYMBOL	UNIT	DESCRIPTION
Thickness	H	Å (Angstrom)	Physical thickness of the layer
Roughness	σ	Å (Angstrom)	RMS roughness (Gaussian σ) of the layer's <i>upper</i> interface; for the top layer it is the surface roughness, for the substrate the roughness of the substrate surface (see Algorithm)
Density	ρ	g/cm ³	Material density

Additionally, each layer has a **material** assignment that links it to an entry in the Henke atomic form factor database.

Density 0 and the substrate density v3.9.1

A density of **0** means “use the bulk density from the Henke table”. This holds for every layer and, since version 3.9.1, for the substrate too: a substrate density you enter is now used in the calculation. Earlier versions always used the bulk density for the substrate and silently ignored the value shown.

A density gradient ([function profile](#) on ρ) takes the layer's entered density as its constant term. With $\rho = 0$ it therefore starts from zero, not from the bulk value, and the layer is empty in the first period: enter an explicit density before adding a density gradient.

So that old results do not change, a project saved by an earlier version opens with its substrate density set to 0 (bulk), which is what those models were always computed with. To use a specific substrate density, for example 2.2 g/cm³ for fused silica or 2.5 for soda-lime glass, enter it and save the project.

Adding a layer

1. Select a stack or model in the structure editor.
2. Choose **Structure > Add Layer** or use the  toolbar.
3. The Layer Properties dialog opens with fields for:
 - **Layer material:** Select or type the name of a material in the materials database (e.g., “Si”, “Mo”, “SiO2”). The name is not parsed as a formula: it must match a Henke table, otherwise the calculation stops with “Material ... not found in the database”. Add new compounds with [Tools > Create New Material](#) first.
 - **H (Å):** Thickness in Angstroms
 - **σ (Å):** Roughness in Angstroms
 - **ρ (g/cm³):** Density in g/cm³

Navigating layers

The Layer Properties dialog includes << and >> buttons to move to the previous or next layer without closing the dialog.

Layer operations (via Structure menu or toolbar)

-  **Insert Layer** — insert before the selected position
-  **Copy Layer** — copy to internal clipboard
-  **Paste Layer** — paste from internal clipboard

-  **Cut Layer** — copy and delete
 -  **Delete Layer** — remove the layer
-

Quick Parameter Editing

In the structure editor, you can adjust layer parameters directly:

- Click a parameter value to edit it in-place
 - Use the **Increment** selector (in the structure toolbar) to set the step size for fine-tuning
 - Changes are applied immediately; if Auto-Calc is enabled, the reflectivity recalculates in real time
-

Undo

Choose **Structure > Undo Structure Changes** to revert the last structural modification.

Model Text (JSON)

For advanced users, **Structure > Edit Model Text** opens a JSON editor showing the raw model definition. This allows direct text editing of the complete structure.

See Also

- [Profile Extensions](#)
- [Tools](#)
- [Fitting > Setting Fit Limits](#)

8 Experimental Data

Loading and pasting measured curves, the XRDML import, the data format, normalize, smooth, trim, the quality assessment, and export

Loading Data from File

1. Select a model or data item in the project tree.
2. Choose **Data > Load ...** or click the  **Load Data** toolbar button.
3. Pick the file. The dialog opens on **XRDML (PANalytical)**; the other filters are ASCII data (`.txt` , `.csv` , `.tet`), counter files (`.dat`) and all files. A text file holds two columns (angle or wavelength, and reflectivity; see [Data Format](#)); an `.xrdml` file is read as described below.
4. The data become a new item under the selected data folder (or the Data root) and appear as a series on the main chart. The header lines at the top of a text file that start with `*` are copied into the item's description.

Loading XRDML Files

A PANalytical / Malvern `.xrdml` file (schema 1.0 to 2.x) is read directly, and the file's own facts are used rather than typed in:

- **Angle axis.** A 2Theta–Omega scan is in 2θ , a rocking curve in ω . The curve is brought to the unit the **2-theta** switch of the calculation settings says, so the chart and the fit see the same angle whichever way the file was scanned.
- **Intensity.** Counts are divided by the counting time (beam attenuation factors, when the file has them, are multiplied in) and the curve is normalised to 1 at its maximum. A zero or negative count is replaced by the smallest positive rate before it (1000 counts/s when none precedes it), as a log chart needs. The peak count rate of the corrected curve (attenuation factors included), the counting time, the detector and the number of floored counts are written into the item's description; the rate the detector itself saw is re-read from the file by [Assess XRR quality](#).
- **Wavelength.** The wavelength the file implies is written into the calculation settings: for a $K\alpha$ measurement through an optic that passes both lines it is the doublet weighted by the file's $K\alpha_2/K\alpha_1$ ratio (1.541874 Å for Cu), for a monochromator or a hybrid mirror it is $K\alpha_1$. A period fitted at $K\alpha_1$ when the doublet was measured comes out 0.08 % short.
- **Source file.** The path of the file is kept in the description as `* Source file:` , and [Assess XRR quality](#) reads the file again for the count rates the chart no longer carries.

Note: the normalisation to 1 at the maximum is a presentation scale, not a measurement: on a well-aligned specimen the maximum is the total-reflection plateau, on one whose plateau is depressed by alignment it is the first Bragg order. The [Normalize](#) step still decides the scale a fit works at.

Pasting Data from Clipboard

1. Copy two-column data from another application (e.g., a spreadsheet).
2. Choose **Data > Paste from Clipboard** or click the  **Paste Data** toolbar button.

3. The data become a new item *Data N.dat* in the Data folder. The same rules as for a text file apply (see [Data Format](#)); a spreadsheet copy is TAB-separated and reads directly.

Data Format

A text file holds one point per line, in two columns:

- Column 1: the angle in degrees, in the unit the **2 θ** switch of the calculation settings shows (2 θ when it is checked, the default; θ when it is not), or the wavelength in Å in wavelength mode. A text file is not converted, and toggling the switch later does not convert data already loaded.
- Column 2: the reflectivity (or an intensity to be normalised afterwards).

The reader is strict:

- The two numbers are separated by **one TAB** (or, in a file without TABs, by one space). A line with a third column, or with several spaces between the numbers, is skipped.
- The first line without a TAB switches the reader to space-separated columns for the rest of the file, so a TAB-separated file must not contain a header line without a TAB.
- The angle must start with a digit; the intensity may also carry a sign. A leading blank or a leading point makes the line skipped. A zero or negative intensity (background-subtracted data) is replaced by the smallest positive value before it, as for an XRDML file. A comma is accepted as the decimal separator.
- Lines that are not two such numbers are skipped. A zero reflectivity is replaced by the smallest positive value before it (1000 when none precedes it).
- Lines at the top that start with * are copied into the item's description. A first line containing **Sample** removes the first 21 lines (the header of the old counter files).

Example (the gaps are single TABs):

```
* Mo/Si multilayer, 2024-01-15
0.100  1.000000
0.200  0.999850
0.300  0.998200
0.500  0.956300
1.000  0.523100
2.000  0.012450
5.000  0.000023
```

Tip: You can also copy and paste table data directly from **OriginLab Origin**. The recommended workflow is to import your raw data into Origin first, then copy-paste the two-column table into X-Ray Calc.

Data Processing

Normalize

 **Data > Normalize** asks for a coefficient and divides every reflectivity value of the active data item by it: 0.099 turns a plateau read at 0.099 into 1.

Data > Normalize (Auto) scales the data so that their maximum lies on the active model's calculated curve: the data are divided by $D_{\max} / R_{\text{model}}$, where $D_{\max}$ is the largest measured value and R_{model} the model's value at the first calculated point at or above its angle. Calculate the model first, over a range that covers the angle of the data maximum; if it does not, the command says so and leaves the data unchanged.

Smooth

 **Data > Smooth** replaces each point by the plain mean of six neighbouring points (three below it, itself and two above), computed on the linear reflectivity. The first three points are left unchanged and the last three take the mean of the last six. On a logarithmic curve this fills Kiessig minima and broadens sharp Bragg peaks, so use it on noisy tails, not on the features the fit reads the structure from. Each click smooths again.

Trim

 **Data > Trim** deletes the data points that lie outside the calculation range (θ_1 – θ_2 , or λ_1 – λ_2 in wavelength mode), at both ends. Set the range first. If no point lies inside it, the data are left unchanged.

Note: Normalize, Smooth and Trim change the data item itself and are saved with it at once; **Undo** covers the structure only. The file you loaded is not touched, so to undo, load it again.

Data Conditioning Walkthrough

Raw experimental data often requires conditioning before fitting. Common issues include a long noisy tail, shadowed regions at low angles, noise, and unnormalized intensity. Here is a typical conditioning workflow:

Step 1 — Normalize. Compare the experimental and calculated intensities at a known point (e.g., the total reflection region). Determine the normalization factor and apply it via **Data > Normalize**. For example, if the experimental intensity at 0.4° is 0.099, divide by 0.099 to bring the curve to unit reflectivity. The fit need not take this factor as final: with **Solve scale in χ^2** (on by default) it solves the scale within a window around it, since a single point rarely fixes the absolute level to better than several per cent (see [Solving the Measured Scale](#)).

Step 2 — Smooth. If the tail is noisy, apply **Data > Smooth**. It is a six-point moving average, so it also rounds the fringes and peaks: use it sparingly (see [Smooth](#)).

Step 3 — Trim. Remove data points outside the useful range:

1. Set the calculation angle range to cover only the reliable portion of the data (e.g., 0.4° to 4.5°).
2. Click **Calc** and note the χ^2 value.
3. Select **Data > Trim** to discard points outside the calculation range.
4. Click **Calc** again — the χ^2 should decrease significantly, confirming that noisy tail data has been removed.

After these steps, the experimental curve is ready for fitting.

Assessing a Measurement

Data > Assess XRR quality ... answers one question before any fitting: is this measured curve worth fitting? It runs eight checks on the active data item and shows one row per check with a verdict (**pass**, **warn**, **fail** or **unknown**), the value, the threshold it was judged against, and a sentence saying why. The same checks are the `assess_xrr` tool of the [MCP server](#).

CHECK	WHAT IT LOOKS AT
Counting	The highest count rate the detector saw, against its linear limit. The rate comes from the <code>.xrdm1</code> behind the item; a curve loaded from a text file has none.
Plateau vs first order	The first Bragg order over the low-angle maximum. Above 1 is impossible for a true reflectivity and fails; with a design, a ratio above what the design's own model gives warns. On a single film or a bare substrate the maximum after the plateau is a Kiessig fringe, and the row says so.
Total reflection	Whether the scan starts below the design's critical angle and rises to a plateau. A curve whose maximum is its first point started on the way down and fails; so does a scan that starts at or above the design's critical angle.
Orders visible	How many of the Bragg orders the design puts inside the range stand above the background, against how many the design's model, scaled to the measured plateau, puts there. Fewer measured than expected is a measurement problem, not a fitting one.
Range below background	The fraction of the points at or below the background, and where the useful range ends: the part of the scan that bought nothing. Reported without a verdict.
Sampling	Points per Kiessig fringe, from the design's total thickness, and per order, from its period. Under two points per fringe the fringes are aliased and no fit will see them (fail); under the number in the window (default 3) it warns.
Footprint	The knee $\text{asin}(\text{beam width} / \text{sample length})$ below which the beam overfills the specimen, and whether the plateau maximum lies below it (warn) or not (pass).
Zeros	How many points have a non-positive count, their fraction, and where they start. Any zero warns; a curve from a text file carries no counts and is unknown.

The background is the median of the last 100 points. An order counts as visible when it stands more than the visible factor (default 3) times above it; the useful range ends at the last point above that level.

The window holds the numbers nobody records in a file: the detector's linear limit in counts per second, the specimen length and the beam width in millimetres, the resolution the design is convolved with (taken from the calculation settings), the factor above the background an order must stand to count as visible, and the minimum points per fringe. Change any of them and click **Assess** to re-run. An empty detector limit, specimen length or beam width leaves its check **unknown**: the program never invents those. An empty visible factor or points-per-fringe returns to 3, an empty resolution to the calculation setting. **Use the current structure as the design** feeds the structure editor's model, including a bare substrate for a glass reference, to the checks that need a period, a thickness or a critical angle.

The text block under the table is one line per check, ready to be filed with the specimen; **Copy summary** puts it on the clipboard. The overall verdict in the window's title is the worst of pass, warn and fail; unknown checks do not count, and the verdict is unknown only when every check is.

When the item came from an `.xrdm1` that is still where it was loaded from, the checks run on the curve as the file holds it: a Normalize, Smooth or Trim done here is not seen. The wavelength is the calculation settings' value in either case.

Tip: the counting and zero-count checks need the raw file. Load the measurement from its `.xrdm1` rather than from a converted text file, and keep the file where it was loaded from: the assessment re-reads it through the `* Source file:` line of the item's description.

Exporting Data

- **Data > Copy to Clipboard** — copies the active data item as TAB-separated text under a three-line header (column names, units, a blank line), the angle to 0.001 and the reflectivity to five significant digits
- **Data > Export to File** — saves the active data item to a text file, in the same layout as Copy to Clipboard (the project tree's **Export Data** does the same). The calculated curve is saved with **Result > Save**.

See Also

- [File Formats > Experimental Data Files](#)
- [Fitting \(Optimization\)](#)
- [Troubleshooting](#)

9 Calculation

Theta and lambda modes, polarization, number of points, auto-calculation, and multi-threading

Algorithm

XRC3 uses the exact Parratt recursion for specular X-ray reflectivity. The medium above the structure is vacuum ($\epsilon = 1$) and the substrate is semi-infinite. The dielectric constant of each layer is $\epsilon = 1 - 2\delta + 2i\beta$, with δ and β proportional to $\rho\lambda^2f_1/A$ and $\rho\lambda^2f_2/A$ (ρ the density, A the molar mass). The atomic scattering factors f_1 , f_2 come from the Henke tables (Henke *et al.*, 1993), linearly interpolated in photon energy, $E = 12398.6/\lambda$ eV (λ in Å).

Interface roughness is Gaussian. The Fresnel coefficient of every interface is multiplied by a Névot–Croce-type factor $\exp(-q_0q'\sigma^2/2)$, where $q_0 = 4\pi \sin\theta/\lambda$ is the wave-vector transfer in vacuum and q' the one refracted across the interface; the same factor is applied to s and p. The σ of a layer is the roughness of its *upper* interface (see [Layers](#)).

Reflectivity can be calculated as a function of either the grazing angle or the wavelength. Calculated values below the chart's minimum-limit setting (the reflectivity floor in the Chart Info bar, default 10^{-7}) are raised to it; the floor applies to the χ^2 too.

Calculation Modes

X-Ray Calc 3 supports two calculation modes:

Theta Mode (by angle)

Calculates reflectivity as a function of the grazing angle (measured from the surface) at a fixed wavelength.

PARAMETER	DESCRIPTION
θ_1	Start angle (degrees)
θ_2	End angle (degrees). The calculated points are $\theta_1 + i\cdot(\theta_2 - \theta_1)/N$, so θ_2 itself is not calculated.
λ (Å)	Fixed wavelength in Angstroms (default: 1.54043 Å, close to Cu $K\alpha_1$, 1.5406 Å)
$\Delta\theta$	Instrumental angular resolution: the FWHM, in degrees θ , of the Gaussian the calculated curve is convolved with (default 0.015; 0 = no convolution). It is not the step. The Gaussian is cut off at $\pm 0.1^\circ$ on the chart axis, so a resolution wider than about 0.08° (0.04° with 2θ ticked) loses its tails. The kernel is normalised, so the convolution keeps the curve's level whatever the step and however much the tails are cut. (Before 3.9.3.1010 it was not, and on a grid coarse against the FWHM the convolved curve was scaled — by 1.9 at 0.01° per point with a 0.005° FWHM.)
2θ	When checked, θ_1 , θ_2 and the chart axis are in 2θ (the calculation divides by 2). $\Delta\theta$ is always entered in θ .

Convolution edges: the convolution needs 0.1° of curve on each side of a point. The first points of the curve (0.1° worth) are left unconvolved and the last ones are replaced by a short trailing moving average; the χ^2 leaves out these end points (see [Understanding the Cost Function](#)). The convolution assumes evenly spaced points.

Lambda Mode (by wavelength)

Calculates reflectivity as a function of wavelength at a fixed grazing angle.

PARAMETER	DESCRIPTION
λ_1	Start wavelength (Angstroms)
λ_2	End wavelength (Angstroms). The step is $(\lambda_2 - \lambda_1)/N$.
θ	Fixed grazing angle in degrees, measured from the surface (90° = normal incidence; default 85°). The 2θ box does not apply here.
$\Delta\lambda$	Spectral resolution: the FWHM ($\text{\AA}$) of the Gaussian the curve is convolved with (default 0 = none; cut off at $\pm 0.1 \text{ \AA}$). It is not the step.

Lambda mode reads the atomic scattering factors f_1 , f_2 at every wavelength of the scan, so the energy dependence of δ and β , absorption edges included, is followed; a function (gradient) profile counts its periods from the surface at every wavelength. (Before 3.9.3 f_1 , f_2 were read once, at λ_1 , and a profile off the top layer was mis-indexed after the first wavelength.) Lambda mode runs on a single thread.

Polarization

- **s-type:** S-polarization (TE mode, electric field perpendicular to the plane of incidence)
- **sp-type:** unpolarized beam, $R = (R_s + R_p)/2$, with the same roughness factor applied to both components

Guidance: What decides is the grazing angle, not the photon energy. Away from the critical angle $R_p \approx R_s \cos^2 2\theta$, so sp is lower than s by the fraction $(1 - \cos^2 2\theta)/2$: under 0.5 % below $\theta \approx 3^\circ$, about 1.5 % at 5° , 6 % at 10° and 20 % at 20° . Use **s-polarization** at small grazing angles, where the difference is negligible and the calculation is faster. Use **sp-polarization** for an unpolarized source at larger angles, for example near-normal EUV and soft X-ray geometries; R_p falls to nearly zero near the Brewster angle, $\theta \approx 45^\circ$.

Number of Points (N)

The **N** parameter sets the number of calculated points (default 2000 in a new session; a project that stores none opens with 1000). More points give smoother curves but take longer to compute. In Theta mode N is rounded down to a multiple of the number of threads.

When a measured curve is linked to the model and the model curve is visible, N is not used: the model is calculated at the measured angles.

Running a Calculation

1. Select a model in the project tree.
2. Set the calculation parameters (mode, angle/wavelength range, polarization, N).
3. Choose **Calc > Calculate** or click the  **Calc Run** toolbar button.
4. The calculated reflectivity curve appears on the main chart.

When a measured curve is linked, **Calculate** also reports the χ^2 against it, and the plain χ^2 , in the Chart Info bar, at the scale shown in the **scale** pane (see [Understanding the Cost Function](#) and [Solving the Measured Scale](#)).

Calculate All (Calc > Calculate All) calculates reflectivity for every model in the project.

Auto-Calculation

Auto-Calc: When enabled in Settings, the reflectivity is recalculated automatically whenever you modify the structure. This provides real-time feedback as you adjust layer parameters.

Multi-Threading

In Theta mode the calculated points are divided into as many contiguous segments as there are threads, and the segments are computed in parallel (with a linked measured curve, its points are divided the same way). Lambda mode runs on a single thread. Configure the number of threads in **File > Settings > Calc & Fit:**

- **Auto (Use all)** — one thread per logical processor available to the process (default)
- **2, 4, 8, 12, 16, 32, 64** — limit to a specific number of threads

Performance: With a 16-core CPU, the typical simulation time for XRR ranges from 0.05 to 0.25 s for a periodic multilayer model consisting of 300 stacks of 4 layers each (3000-point curve). This speed enables real-time manual curve fitting.

See Also

- [Defining a Model](#)
- [Charts & Visualization](#)
- [Application Settings](#)

10 Fitting (Optimization)

The Shake LFPSO algorithm, fit limits, fitting modes, parameters, convergence, and the cost function

Fitting adjusts model parameters to minimize the difference between calculated and experimental reflectivity curves. X-Ray Calc 3 uses the **Shake LFPSO (Lévy Flight Particle Swarm Optimization)** algorithm — a population-based global optimization method with an enhanced “Shake” mechanism to escape local minima (Li *et al.*, 2023; Penkov *et al.*, 2024).

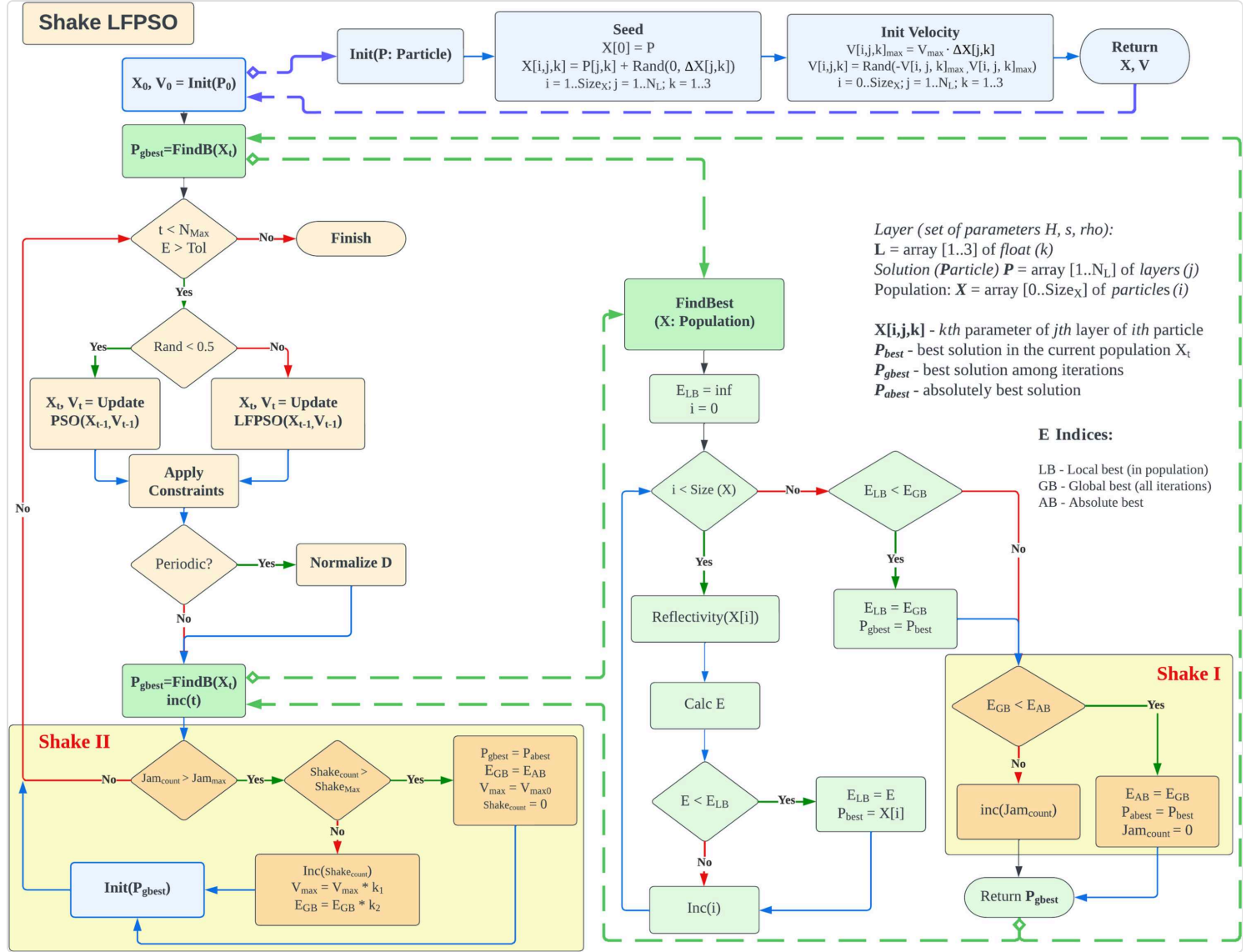


Figure 3. General diagram of the Shake LFPSO algorithm. Bold dashed lines indicate calls to subroutines. (From Penkov *et al.*, 2024.)

The LFPSO algorithm creates a population of solutions (particles), each being a vector of structural parameters. The population is iteratively modified to produce better-fitting solutions. The Lévy flight is a random walk that uses the Lévy distribution to determine step sizes, providing efficient global exploration due to occasional long jumps. The Shake modification re-initializes the population when convergence stalls, helping the algorithm escape local minima.

Version 3.6 introduces several [convergence enhancements](#) that improve exploration/exploitation balance, boundary handling, and shake triggering. Six improvements are always active; the Clerc-Kennedy constriction factor is optional (enabled by default in Advanced Settings).

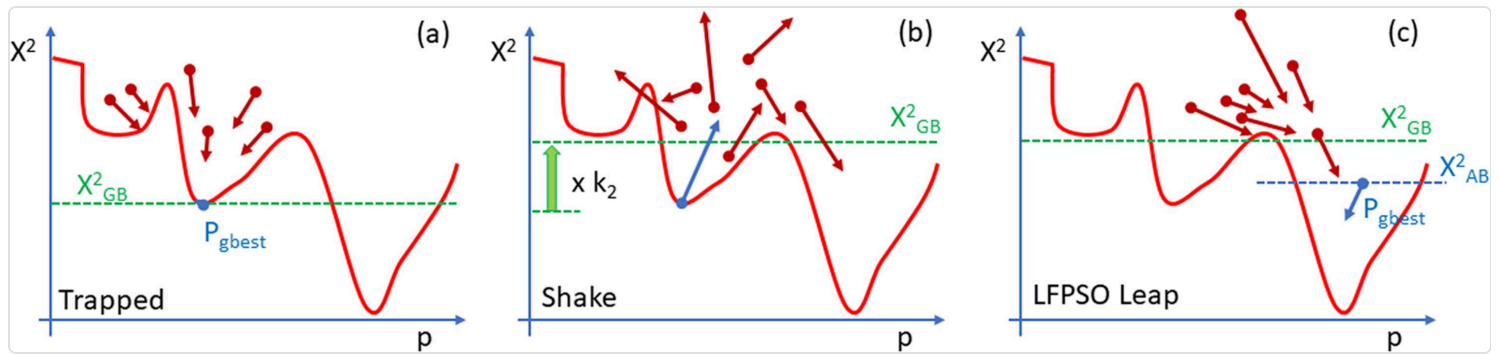


Figure 4. The Shake concept. (a) All particles trapped in a local minimum. (b) Shaking event: locations and velocities are randomized based on P_{gbest} ; V_{max} is multiplied by k_1 and the cost function threshold is lifted by k_2 . (c) After LFPSO leap, a new best solution is found. (From Penkov et al., 2024.) The Advanced Settings dialog labels these coefficients the other way round: there k_1 lifts the threshold and k_2 multiplies V_{max} (see [Advanced Fitting Settings](#)).

Prerequisites for Fitting

Before starting a fit, you need:

1. **A defined model** with initial parameter estimates
2. **Experimental data** loaded into the project, linked to the model and visible on the chart (otherwise the fit stops with *Measured curve is not linked!*)
3. **Fit limits** set for each adjustable parameter

Setting Fit Limits

1. Select the model in the project tree.
2. Click the **Set Fit Limits** button in the structure toolbar.
3. The Limits dialog opens, showing a table of all layer parameters (H , σ , ρ), each with a **Fix** column and columns for Min and Max.
4. Click any Min or Max cell to edit its value directly.
5. Only parameters with $\text{Min} \neq \text{Max}$ will be adjusted during fitting. Setting $\text{Min} = \text{Max}$ pins a parameter permanently; to hold one temporarily, freeze it instead (see below).

Freezing Parameters v3.8.2

Freezing holds a parameter at its current value for the next fit *without discarding its limits*. This is the difference between freezing and typing $\text{Min} = \text{Max}$: a frozen parameter keeps the search window you set, so thawing restores exactly that window, and the value it holds follows the model — it tracks the result of every later fit and every manual edit.

- Click a cell in a parameter's **Fix** column to freeze or thaw it. Frozen parameters show **X** and their limits are greyed.
- Select one or more rows and use **Freeze** or **Thaw** to act on whole layers.

- Right-click for **Freeze this stack**, **Freeze this layer**, **Freeze all** and **Thaw all**.
- Each stack's group header counts what is held — *Period stack* — *all 6 frozen* or — *1 of 6 frozen*.

A stack can also be frozen without opening this dialog: right-click it in the structure panel and choose **Freeze stack** or **Thaw stack**. The stack header then shows *[frozen]* or *[3/6 frozen]*, and the two views always agree.

Freeze state is saved in the project file, so it survives closing and reopening. Projects written by earlier versions load with nothing frozen.

Frozen parameters are left alone by **Initialize**, **Narrow**, **Widen** and **Fix**, and are not reported by validation — a frozen parameter with a stale range cannot block a fit.

Limit Actions

The dialog provides action buttons for batch operations on limits:

BUTTON	ACTION
Initialize	Generates limit ranges around current parameter values using the ΔH , $\Delta\sigma$, $\Delta\rho$ fractions (0.25 = $\pm 25\%$): $\text{Min} = V(1 - \Delta)$, $\text{Max} = V(1 + \Delta)$. For density (ρ), uses the Henke database bulk density as the center value when available.
Narrow	Shrinks all unlocked limits toward the current fitted value by 50%, reducing the search space. Useful after a successful fit to refine results.
Widen	Expands limits only for parameters where the fitted value sits at a boundary (within 1% of the range edge). Useful when a fit converges to a limit, suggesting the true optimum may lie outside the current range.
Fix	Automatically resolves all detectable problems: swaps inverted Min/Max, adjusts limits to include out-of-range values, widens at-boundary parameters, enforces physics constraints, and applies geometry coupling. Despite the name, this is unrelated to the Fix columns — it repairs limits, it does not freeze anything.
Freeze / Thaw	Holds or releases every parameter of the selected layers. See Freezing Parameters .

Validation and Visual Feedback

The dialog validates limits in real time and highlights problematic cells:

- **Red cells** indicate **errors** that must be fixed before fitting:
 - $\text{Min} > \text{Max}$ (inverted range)
 - Negative Min for H , σ , or ρ
- **Yellow cells** indicate **warnings** — fitting is allowed but results may be suboptimal:
 - Fitted value below Min or above Max
 - Fitted value at a limit boundary (within 1% of range)
 - Range too wide ($>100\times$) or too narrow ($<1\%$ of value)
 - ρ Max exceeds densest stable element (23 g/cm^3)
 - All parameters locked — nothing to fit

Click **Fix** to resolve all highlighted issues automatically, or address them manually by editing individual cells.

Physics Constraints

All limit operations automatically enforce physical constraints:

- $H, \sigma, \rho \geq 0$ (no negative values)
- $\rho \leq 23 \text{ g/cm}^3$ (densest stable element, Osmium)
- $\sigma_{\text{Max}} \leq \min(\text{own } H, \text{neighboring layer } H)$ — roughness cannot exceed layer thickness or the thickness of the layer below

Fitting Modes

The software provides several structure-representation methods. The user chooses the most suitable depending on the nature of the coating:

STRUCTURE REPRESENTATION	MODE	APPLICABILITY	COMPUTATION TIME
Non-periodic coating	Irregular	Regular non-periodic coatings	Long
Periodic	Periodic	Periodic multilayer mirrors (PMMs)	Short
Periodic with arbitrary deviations	Irregular, applied to a periodic stack	Supermirrors, PMMs with arbitrary distributions	Long
Polynomial distributions	Polynomial	PMMs with gradual change of parameters	Moderate

The Fitting panel offers three modes: **Irregular**, **Periodic** and **Polynomial**.

Irregular Mode

Every parameter of each layer is treated independently. This mode is used for non-periodic coatings consisting of several layers. At the start of fitting, the software expands each periodic stack into individual layers. Once fitting is complete, distribution charts are generated for each parameter of every material type (thickness, roughness, density).

- Each layer has its own fitted thickness, roughness, and density.
- With **Smooth** checked, each particle's period-to-period profile of every parameter is smoothed with a moving average over the **Smoothing window** (Advanced Settings) after every update. This is a constraint during the search, not a smoothing of the result; it applies to stacks with $N > 1$ and to parameters that are not paired.
- Specific parameters can be “stack-locked” (paired) — e.g. roughness and density remain constant while thickness varies across the stack.

Example: A Mo/Si PMM with 60 repeated stacks would require 726 variables if treated individually. Using the periodic mode, this reduces to just 18 variables.

Periodic Mode

The coating consists of repeated stacks; each stack consists of several layers. Parameters are fitted per-layer within one period, then applied to all periods. After every move a periodicity keeper rescales the layer thicknesses of each stack with $N > 1$, in proportion to each layer's thickness and within its bounds, so that the period D keeps its value.

The period is fixed. A Periodic fit in the program holds the period D of every repeating stack at its value in the start model: it changes the ratio of the layer thicknesses (and σ , ρ), not D , so it cannot move the Bragg peaks. Set the period first — the peak positions must already match — or use Irregular or Polynomial mode to let it move. (The MCP server can free the period; see [MCP Server > Fitting](#).)

- Suitable for structures with consistent periodicity.
- Dramatically reduces the number of fitting variables for PMMs.

Polynomial Mode

Arbitrary distributions of structural parameters across the stack can be described by polynomial functions of the period number. Instead of fitting N individual layer thicknesses, the software fits just a few polynomial coefficients (a_1 – a_k) plus the base value x_0 :

$$x(j) = x_0 + a_1(j - 1) + a_2(j - 1)^2 + \dots + a_k(j - 1)^k, j = 1, 2, \dots \text{ the period number, } x_0 \text{ the first period's value.}$$

Only repeating stacks ($N > 1$) and parameters that are not paired get coefficients; the others keep a single value. The period is not held, so the Bragg peaks can move. Keep the order low (≤ 5 in practice): high powers of the period number grow very fast, and the order field is not range-checked.

- **Order:** Maximum polynomial order (1 = linear, 2 = quadratic, etc.)
- Useful for smooth, continuous parameter gradients (e.g. graded-period supermirrors, PMMs with unstable deposition rate).

Basic Fitting Parameters

PARAMETER	DESCRIPTION	DEFAULT
Iterations	Maximum number of optimization iterations	100
Population	Number of particles (candidate solutions) in the swarm	100 (a project from a version that did not store it opens with 1000)
Shake	Enable random perturbation to escape local minima	Checked
SeedR	Start the swarm spread uniformly over each parameter's [Min, Max] instead of scattered around the start model ($\pm Ks_{xr}$ of the range). Applies only at the start of a fit; no effect in Polynomial mode	Checked (unchecked for a project that did not store it)
Smooth	Irregular mode only: keep each parameter's period-to-period profile smooth (moving average over the Smoothing window) during the search	Unchecked
PW χ^2	Point-weighted chi-square calculation	Checked
TW χ^2	Theta-weighted chi-square function	None
Solve scale in χ^2	Score every candidate structure at the measured scale that suits it best, within the window, instead of at the anchored scale (see Solving the Measured Scale)	Checked
Window	How far the solved scale may move from the anchored one, as a fraction: the scale stays between $1/(1 + w)$ and $(1 + w)$ times the anchor; 0 pins it	0.2

Theta-Weighted Chi-Square (TW χ^2)

Controls how the chi-square metric weights different angular regions. θ here is the value on the chart axis — 2θ when the 2θ box is checked — so the absolute χ^2 with a TW weight differs by a constant factor between the two axes:

OPTION	EFFECT
None	Equal weight across all angles
sqr	Weight $\propto \theta^2$ (emphasizes high angles)
line	Weight $\propto \theta$
sqrt	Weight $\propto \sqrt{\theta}$
1/sqr	Weight $\propto 1/\theta^2$ (emphasizes low angles)
1/sqrt	Weight $\propto 1/\sqrt{\theta}$

Advanced Fitting Settings

Click the **Advanced Settings** button in the Fitting panel to open the detailed LFPSO configuration dialog.

General

PARAMETER	DESCRIPTION	DEFAULT
Tolerance	Fitting stops when the best weighted χ^2 (as shown, with the PW, TW and scale settings) falls below this value	0.005
Constriction factor	Use Clerc-Kennedy constriction coefficient ($\chi \approx 0.73$) instead of linear inertia decay. Analytically guarantees convergence. See details below .	Checked

LFPSO Parameters

PARAMETER	DESCRIPTION	DEFAULT
SHmax	Shakes do not stop the fit. The first SHmax + 1 shakes each lift the threshold (k_1) and enlarge $V_{\max}$ and Ksxr (k_2); the next one instead returns the swarm to the best solution found so far and restores $V_{\max}$ and Ksxr, and the cycle repeats	3
Jmax	A shake is due once more than Jmax iterations since the start or the last shake have failed to improve the current best (not necessarily consecutive ones)	1
Vmax	Largest step a particle may take per iteration, as a fraction of each parameter's range (Max – Min); velocities are clipped to $\pm V_{\max}$	0.3 (0.1 for a project that did not store it)
ω_1	Constant part of the inertia weight $\omega = \omega_1 + \omega_2(1 - t/T)$. Ignored when Constriction factor is on	0.3
ω_2	Decaying part of the inertia weight: ω falls from $\omega_1 + \omega_2$ to ω_1 over the run. Ignored when Constriction factor is on	0.3
k_1	Shake lift of the best cost: at a shake the current best χ^2 threshold is multiplied by k_1 , so the new swarm can accept worse solutions	1.41
k_2	Shake enlargement: at a shake $V_{\max}$ and Ksxr are multiplied by k_2	1.41
Ad.V	Scale the attraction coefficients c_1 , c_2 with the same decaying schedule as the inertia, $\omega_1 + \omega_2(1 - t/T)$, instead of 1. Ignored when Constriction factor is on	Unchecked
Ksxr	Re-seeding spread, in all modes: after a shake (and at the start when SeedR is off, always in Polynomial mode) particles are scattered within $\pm Ksxr$ of each parameter's range around the best solution	0.2

The coefficient the paper calls k_1 is the dialog's k_2 and vice versa (see Figure 4).

Irregular Mode Settings

PARAMETER	DESCRIPTION	DEFAULT
Smoothing window	Number of neighbouring periods averaged when Smooth is checked (a window W averages W + 1 consecutive periods); -1 = automatic, one tenth of the periods (at least 1)	-1

Polynomial Mode Settings

PARAMETER	DESCRIPTION	DEFAULT
Polynomial factor	Divisor per order: the order-p coefficient is searched within (Max - Min)/F ^p with velocity limit V _{max} /F ^p (1-15)	10

Running a Fit

1. Ensure a model with experimental data and fit limits is selected.
2. Configure fitting mode and parameters.
3. Choose **Calc > Auto Fitting** or click the  **Fast Forward** toolbar button.
4. The fitting process begins:
 - The main chart updates in real time, showing the best-fit curve converging toward the experimental data.
 - The **Fitting Progress** chart page shows the χ^2 convergence curve.
 - The status bar displays elapsed fitting time, and at the end the device the fit ran on (see [GPU Acceleration](#)).
 - The **scale** pane of the Chart Info bar reads *solved* or *anchored*; when the fit ends it shows the solved ratio (see [Solving the Measured Scale](#)).
 - If **Live Update** is enabled (in Settings), the structure editor updates to show current best-fit parameters.
5. Fitting stops when:
 - The maximum number of iterations is reached, OR
 - The tolerance threshold is met, OR
 - You click the **Stop** button.

GPU Acceleration v3.9.0

Each iteration of the fit scores every particle of the swarm, and almost all of that time goes into the Parratt recursion over every layer at every measured angle. When the computer has a graphics card, X-Ray Calc 3 hands the whole swarm to it in one piece: every particle at every angle is computed at once, together with the resolution convolution and the χ^2 . The models themselves are still built on the CPU, in parallel.

- On a modern graphics card an iteration runs one to two orders of magnitude faster than on all CPU cores (for a 150-period W/B₄C multilayer measured at 2000 points, about 70× with 500 particles and 170× with 5000, against all 32 threads of a Ryzen 9 9950X3D on an RTX 5080). That makes much larger swarms practical.

- Small models gain less: with a few layers and a few hundred points the fixed cost of each iteration dominates.
- The GPU computes in single precision, like the CPU engine, and its accuracy against a double-precision reference is at least as good. Its χ^2 agrees with the CPU's well within 1 %, so the search behaves the same; it follows a different path through the swarm, so a GPU fit and a CPU fit with the same seed do not end on identical numbers.
- The χ^2 and the curve reported at the end of a fit are always recomputed on the CPU, so they are exactly what **Calc** gives for the fitted model.
- Wavelength scans are fitted on the CPU. If the graphics card is missing or fails, the fit continues on the CPU.

The option is **Use the GPU for fitting** in [Settings > Calc & Fit](#), on by default. When a fit ends, the status bar shows how long it took and the device it ran on (for example *Fitting time: 2.14 s on NVIDIA GeForce RTX 5080*), and **Export fit results** records them as `fittingSeconds` and `fittingDevice`. Any graphics card with Direct3D 11 works; nothing extra needs installing.

Staged Fitting v3.8.2

A complex model rarely fits well in one pass. Staged fitting lets you settle part of it, hold that part still, and concentrate the search on what remains — typically fitting the periodic stack first, then freezing it and working on the top and bottom stacks with a different weighting.

1. Run a fit as usual and let it converge.
2. Freeze the part that has settled — right-click the stack in the structure panel, or use the **Fix** columns in the Limits dialog.
3. Change whatever you want for the next pass, the weight type above all.
4. Choose **Calc > Resume Fitting** or click the  **Resume Fitting** toolbar button.

Resume Fitting is available once a fit has produced a result, and differs from **Auto Fitting** in three ways:

- **It re-centres the search.** Every free parameter's window slides so the value the last fit reached sits at its centre, keeping the width you set. A parameter that ended hard against a boundary can therefore carry on past it, instead of being stuck at the wall.
- **It keeps your fit extensions** without asking. A resumed fit continues from the depth profiles they hold, so the stale-extension prompt would be meaningless there.
- **It continues the convergence chart** rather than clearing it, so both passes are visible on one axis.

The best χ^2 is *not* carried across a resume. Changing the weight type changes what is being minimised, so the two passes' numbers are not on the same scale and comparing them would be misleading.

In Polynomial mode a frozen stack keeps its depth profile: the fit is seeded from the gradient the previous pass produced, so freezing holds the grading rather than flattening it.

Freezing everything leaves nothing to fit, and validation says so.

Convergence Enhancements v3.6

Version 3.6 introduces several improvements to the LPSO algorithm that significantly improve convergence speed and reliability. Six enhancements are always active and require no configuration; one is optional.

Always-On Improvements

ENHANCEMENT	DESCRIPTION
Reflective boundaries	When a particle exceeds the search domain, it bounces back into the valid range (like a ball hitting a wall) with 0.5× velocity damping, instead of being clamped at the boundary. This prevents particles from “sticking” at limits and keeps them active in the search. (Irregular and Periodic modes; in Polynomial mode x_0 is clamped, and higher-order coefficients are zeroed from the top order down until the profile fits within the limits.)
Adaptive Lévy scale	The Lévy flight step size starts large (0.10) for broad exploration and decreases to 0.01 by the end of fitting for fine-tuning. Previously, a fixed value of 0.01 was used throughout.
Adaptive PSO/Lévy switching	Each iteration the whole swarm takes either a Lévy or a PSO step, chosen by one random draw. The Lévy probability falls from 70% to 30% over the run and rises by 5 points for every non-improving iteration since the last shake (at most 90%). Previously, a fixed 50/50 split was used.
Random-peer Lévy targets	30% of Lévy flights are directed toward a randomly chosen peer particle instead of the global best. This creates diverse search directions and helps explore regions that global-best targeting would miss — particularly valuable for the multimodal fitness landscapes common in X-ray reflectivity.
Population diversity monitoring	The algorithm now tracks the normalized spread of particle positions across all parameters, providing a measure of how converged or diverse the swarm is.
Diversity-aware shake	The shake mechanism now checks population diversity before triggering. If the swarm is still diverse (spread $\geq$ 1% of parameter ranges), the shake is delayed for up to 3 additional iterations to let ongoing exploration proceed. If diversity has collapsed ($<$ 1%), the shake fires immediately. This prevents premature reinitializations that waste computation.

Constriction Factor (Optional)

When the **Constriction factor** checkbox is enabled in Advanced Settings, the standard linear inertia weight decay is replaced by the Clerc-Kennedy constriction coefficient:

$$\mathbf{V} = \chi \times (\mathbf{V} + \varphi_1 \mathbf{r}_1 (\mathbf{p}_{best} - \mathbf{X}) + \varphi_2 \mathbf{r}_2 (\mathbf{g}_{best} - \mathbf{X}))$$

where $\varphi = \varphi_1 + \varphi_2 = 4.1$, $\chi = 2 / |2 - \varphi - \sqrt{\varphi^2 - 4\varphi}| \approx 0.7298$

p_{best} is the best particle of the current iteration: the swarm keeps no per-particle memory. In a Lévy iteration the previous velocity is replaced by the Lévy step, aimed at g_{best} or, 30% of the time, at a random peer.

Unlike the empirical linear schedule, the constriction coefficient is mathematically proven to guarantee convergence while maintaining effective exploration. The higher cognitive and social coefficients ($\varphi_1 = \varphi_2 = 2.05$ vs. the previous $c_1 = c_2 = 1$) give particles stronger attraction to good solutions, while the constriction factor prevents velocity explosion.

When to use: The constriction factor is enabled by default and is recommended for most problems. Consider disabling it if you notice premature convergence on problems with many sharp local minima — the linear inertia decay may provide more aggressive exploration in such cases. When disabled, the ω_1 and ω_2 parameters in the LFPSO section control the inertia schedule.

Understanding the Cost Function

The cost function, shown as χ^2 , measures the deviation between the calculated and the measured XRR curve. It is the weighted mean of the squared *relative* residuals of the logarithms, scaled by 1000:

$$\chi^2 = (1000 / (N - 1)) \times \sum_i [(\log I_{\text{exp},i} - \log I_{\text{calc},i}) / \log I_{\text{calc},i}]^2 \times \omega_{\text{peak},i} \times \omega_{\theta,i}$$

where $I_{\text{exp},i}$ and $I_{\text{calc},i}$ are the measured and computed intensities at the i -th measured angle, N is the number of measured points, ω_{peak} is the peak weight and ω_{θ} the incidence-angle weight. The sum runs over every measured point except half the resolution-convolution window at each end (the points within about 0.1° of it), which the convolution cannot compute in full. Without a resolution every point is summed. (Before 3.9.3.1010 one further point was dropped at the end, whether or not there was a convolution.) The calculated curve is floored at the lower limit of the chart's reflectivity axis ($1\text{E-}7$ by default), so that setting also affects χ^2 wherever the data fall below it. With [Solve scale in \$\chi^2\$](#) checked, $\log I_{\text{exp},i}$ carries the solved offset a .

Peak Weight Function (PW χ^2)

Specifically designed for periodic multilayer mirrors. Parameters of the periodic stack primarily affect the intensity of primary diffraction peaks, while Kiessig fringes are affected by the substrate and surface layers. The peak weight concentrates the fit on the primary features of the curve. The measured curve is averaged over a moving window of 5 % of its points (stored in the project, not editable in the program; the first half-window of points always gets weight 1); a point that stands more than three times above that average — a Bragg peak — gets the weight $\omega_{\text{peak}} = I_{\text{exp}} / \text{average}$, every other point the weight 1. Unchecked, $\omega_{\text{peak}} = 1$ everywhere.

Incidence-Angle Weight Function (TW χ^2)

Allows concentration of the fitting process on distinct segments of the XRR curve by weighting different angular regions differently: ω_{θ} is 1, θ^2 , θ , $\sqrt{\theta}$, $1/\theta^2$ or $1/\sqrt{\theta}$ as set in the [TW \$\chi^2\$ list](#).

Plain χ^2

The Chart Info bar also shows the **plain χ^2** : the same sum with both weights set to 1. It is never minimised; it reports the bare disagreement between data and model on the same scale as the χ^2 , so that fits made with different weight settings can still be compared.

Key properties of the cost function:

- **Lower is better** — $\chi^2 = 0$ means a perfect fit
- Displayed in the Chart Info bar as both current and best values, beside the scale it was taken at
- Calculated on the logarithm of the intensity, so that the reflectivity's fall over many decades does not leave the high-angle part of the curve without weight
- Each residual is relative to $\log I_{\text{calc}}$: the points where the reflectivity is close to 1, near and below the critical angle, weigh the most
- The value depends on the weights, so χ^2 values are comparable only between fits made with the same PW, TW and scale settings; the plain χ^2 removes the weights but not the scale

Solving the Measured Scale

A reflectometer does not measure reflectivity. It counts reflected photons, and the reflectivity is that count divided by the intensity of the incident beam, $R(\theta) = I(\theta) / I_0$. I_0 is seldom known to better than several per cent: it comes from a separate direct-beam scan through attenuators whose factors are themselves calibrated, the source drifts between that scan and the measurement, and the detector loses counts at high rates. What reaches the detector at small angles also depends on the specimen:

- **Footprint.** A beam of height b illuminates a length $b / \sin \theta$ of the specimen. Below the angle at which that length equals the specimen's, part of the beam misses the specimen, and the measured plateau of total external reflection is lower than the true one.
- **Alignment and curvature.** An error in the ω zero, or a bowed substrate, lowers and reshapes the region around the critical angle, where the curve is steepest.
- **Normalisation.** [Normalize](#) sets the scale from a single point — in the automatic variant the largest measured intensity of the curve, matched to the model at that angle. The anchor inherits whatever error that one point carries, and depends on the model it was matched to.

All of these multiply the whole curve by one factor s : $I_{\text{meas}}(\theta) = s \cdot R(\theta)$. On the logarithmic scale the fit works on, a factor is a constant offset, $\log I_{\text{meas}} = \log R + \log s$: the measured curve moves up or down without changing shape. That does not move a Bragg peak or change the spacing of the Kiessig fringes, which the thicknesses set, but it changes every absolute intensity — the height of the plateau, the reflectivity of the Bragg peaks, the rate at which the curve falls — and those are what the fit reads roughness and density from. A fit held at a wrong scale reaches for the parameters that can raise or lower the whole curve: a scale that is too low is typically absorbed as extra roughness, one that is too high as extra density contrast. The structure comes out biased, and its χ^2 is higher than the structure deserves.

What the option does

With **Solve scale in χ^2** checked, the scale is treated as a *nuisance parameter*: a quantity the χ^2 needs but which is not itself a result. Every candidate structure the swarm tries is scored at the scale that suits it best. Write $a = \log_{10}(s / s_{\text{anchor}})$ for the offset from the anchored scale, and d for a point's residual $\log I_{\text{meas}} - \log I_{\text{calc}}$. The offset turns every residual into $d + a$, and because the χ^2 is a weighted sum of squared residuals it is a parabola in a :

$$\chi^2(a) = S_2 + 2a S_1 + a^2 S_0, \quad a^* = -S_1 / S_0$$

where S_0 , S_1 and S_2 are the sums over the points of W , Wd and Wd^2 , W being the weight the χ^2 gives a point: the PW and TW weights, and the factor $1/(\log I_{\text{calc}})^2$ that makes every residual a relative one. The best offset a^* is minus the weighted mean residual — the shift that centres the residuals on zero. It is found exactly, in the same pass that computes the χ^2 , so the option adds no dimension to the swarm and costs no time. Every candidate is scored at its own a^* : the scale is *profiled out* rather than fitted, and the swarm searches the structure alone.

The weight explains why the scale matters so much. Because each residual is divided by $\log I_{\text{calc}}$, the points near and below the critical angle, where the reflectivity is close to 1 and its logarithm close to 0, weigh the most — and they are also the points that footprint and alignment errors affect most.

The window

A free scale is not harmless. A structure with the wrong density throughout, or a wrong top layer, can look acceptable if the scale is allowed to carry the whole curve to it: the scale would then absorb physics. The **Window** w says how far you trust the normalisation. The offset is held within $\pm \log_{10}(1 + w)$, so the solved scale lies between $1/(1 + w)$ and $(1 + w)$ times the anchored one: the default 0.2 allows a factor of 1.2 either way. A window of 0 pins the scale to the anchor and gives the same χ^2 as unchecking the option.

- Widen it when the absolute level is poorly known: no direct-beam measurement, a specimen shorter than the beam footprint at the lowest angles, a curve normalised against a start model far from the answer.
- Narrow it when I_0 was measured carefully and the specimen is larger than the footprint: the window then keeps the scale from covering for a wrong model.

Reading the result

- The **scale** pane of the Chart Info bar shows the scale the χ^2 beside it was taken at. *solved* $\times 1.0831$ means the measured curve fits best at 1.0831 times its anchored scale, i.e. about 8 % higher; *anchored* means the option is off. **Calc** shows it for the model on screen, and at the end of a fit for the fitted model.
- *at bound* means the best scale lies outside the window and was held at its edge. Either the normalisation is off by more than the window allows — renormalise, or widen the window if you have a reason to — or the model cannot reproduce the curve's absolute level and the scale was being asked to make up for it. Look at the model before using the numbers of a fit that ends at the bound.
- A ratio well away from 1 but inside the window says the anchor is poor. Applying the ratio with [Data > Normalize](#) and fitting again brings the anchored and the solved numbers together.
- The chart and the saved data stay at the anchored scale: the measured curve is not redrawn at the solved one, so the model can sit visibly above or below the data by the ratio while the χ^2 counts it at the solved scale. The plain χ^2 is taken at the same scale as the χ^2 .
- A solved χ^2 is never higher than the anchored χ^2 of the same structure, since $a = 0$ is always inside the window, so the two kinds of number cannot be compared with each other. Compare fits made with the same setting; uncheck the option to reproduce the numbers of versions that did not have it.

The option and the window are saved with the project; a project saved before the option existed opens with it checked at 0.2. **Export fit results** records them as `solveScale` and `scaleSolveWindow` under `fitParams`,

together with `chi2Scale` (`"solved"` or `"anchored"`), `scaleRatio` and `scaleClamped` . The MCP server's `scale_solve` and `scale_solve_window` are the same option with the same defaults: a fit made in the program and one made through the server on the same settings are the same fit.

See Also

- [Experimental Data](#)
- [Results & Export](#)
- [Illustrative Examples](#)
- [MCP Server > Fitting](#)

11 Results & Export

Saving results, the clipboard, plot and structure images, and auto-saved fits

Saving Results

 **Result > Save Results** saves the calculated reflectivity data to a text file with columns for angle/wavelength and reflectivity.

Copying to Clipboard

-  **Result > Copy to Clipboard** — copy reflectivity data as tab-separated text
-  **Result > Copy as BMP** — copy the chart as a bitmap image
- **Result > Copy as WMF** — copy the chart as a Windows Metafile (vector format, suitable for publications)

Saving the Plot

Result > Save Plot as File saves the chart as an image file.

Structure Image

Structure > Copy as Image copies the visual structure diagram (the layer/stack representation) to the clipboard as an image.

Auto-Save Results

When enabled in Settings (**Automatically save results to output folder after fitting**), fitting results are automatically saved to the configured Output directory after each fitting run completes.

See Also

- [Fitting \(Optimization\)](#)
- [Charts & Visualization](#)

12 Charts & Visualization

The reflectivity chart, zoom and pan, cursor tracking, chart pages, and peak analysis

Main Reflectivity Chart

The large chart in the working area displays:

- **Calculated reflectivity curves** (one per model, color-coded)
- **Experimental data** (if loaded)
- **Best-fit curve** (during and after fitting)

Interaction

- **Zoom:** Click and drag to select a zoom region
- **Pan:** Right-click and drag to pan
- **Reset zoom:** Double-click to reset to full view
- **Scale toggle:** Click the **Scale** button to switch between linear and logarithmic Y-axis
- **Min limit:** Set a cutoff threshold — reflectivity values below this limit are clipped

Cursor Tracking

The Chart Info bar shows the current cursor position (X, Y coordinates) as you move the mouse over the chart.

Chart Pages

Below the main chart are tabbed sub-charts:

Thickness (H)

Bar chart showing the thickness of each layer in the structure. After fitting, this shows how layer thicknesses vary across the structure.

Roughness (σ)

Bar chart showing the interface roughness of each layer. Useful for identifying roughness trends in periodic structures.

Density (ρ)

Bar chart showing the density of each layer, with the applied density profile overlaid.

Profile

Electron density (or refractive index) profile as a function of depth into the structure. This gives a physical picture of the complete multilayer structure.

Fitting Progress

During fitting, this chart shows the χ^2 value vs. iteration number. Use this to monitor convergence:

- A steadily decreasing curve indicates good convergence
 - Plateaus may indicate local minima (shake/re-initialization helps escape these)
 - Sudden jumps up indicate a shake event
-

Peak Analysis

The Chart Info bar includes peak detection and analysis:

- **Peak Max:** Maximum reflectivity value and its position
- **Area:** Integrated area under the peak (useful for quantitative analysis)
- **Period:** Estimated multilayer period from peak spacing

See Also

- [Calculation](#)
- [Results & Export](#)

13 Profile Extensions

Function and table profiles that vary a parameter across a periodic structure

Profile extensions let layer parameters (H , σ , ρ) vary from period to period of a periodic stack — for example a thickness that drifts during deposition. Both kinds of profile are indexed by the period number n , counting from the surface ($n = 1$ is the period nearest the surface), not by depth.

Adding a Profile Extension

1. Select a model in the project tree.
2. Click the  **Add Extension** toolbar button or choose the extension option from the project toolbar.
3. The Extension Type Selector dialog opens. **Function** adds a function profile; the other choices are not available in this version (a table profile is created by a fit, see below).

Profile Types

Function Profile

A function profile varies one parameter (H , σ or ρ) of one layer of a periodic stack as a polynomial in the period number n :

$$v(n) = c_0 + c_1(n - 1) + c_2(n - 1)^2 + \dots + c_K(n - 1)^K, \quad K \leq 9$$

c_0 is always the layer's own value in the structure, so the period nearest the surface keeps the value you entered; you set $c_1 \dots c_K$ in the Profile Function editor, which previews the resulting profile. A new gradient starts with order 1 and $c_1 = 0.14$. The values are not clamped, so a steep polynomial can drive a thickness, roughness or density negative. Other function forms are not implemented in this version: the function list in the editor is fixed to the polynomial.

A density gradient on a layer with $\rho = 0$ starts from zero, not from the bulk density (see [Layers](#)). The [Polynomial fitting mode](#) fits these coefficients.

Table Profile

A table profile holds an individual value of H , σ and ρ for every period of each periodic stack (period 1 = nearest the surface). It is created by an **Irregular** fit of a periodic structure and shows the fitted values per period as a table and a step chart, which you can copy or save; the values cannot be edited in the table. A model has at most one table profile.

The table is used in calculations only while the Fitting mode is **Irregular** or **Polynomial**; in **Periodic** mode every period uses the stack's own values.

Enabling / Disabling Extensions

Each extension can be individually enabled or disabled without deleting it. This lets you quickly compare results with and without a particular profile applied.

See Also

- [Defining a Model > Stacks](#)
- [Fitting > Fitting Modes](#)

14 Tools

Creating materials, editing Henke tables, and the benchmark

Create New Material

 **Tools > Create New Material** opens a dialog to add a custom material to the Henke database.

Choose the number of components, pick each one (an element or an existing material) and give its concentration in atomic per cent, then a name and a density (g/cm^3). The program writes `Henke\<Name>.bin` with f_1 , f_2 and the molar mass averaged per atom with the atomic fractions, $f = \sum c_i f_i$, on the energy grid of the first component (the others are interpolated onto it). The density you enter becomes the bulk density used when a layer's density is 0.

Note: the percentages must add up to 100 at.%; the dialog refuses to save otherwise, and asks before replacing an existing material of the same name.

Edit Henke Table

 **Tools > Edit Henke Table** opens the Henke Table Editor, which allows you to view and modify the atomic scattering factor tables used for reflectivity calculations.

A Henke table holds the atomic scattering factors f_1 and f_2 of one material against photon energy E (the λ column is $12398.6 / E$, in Å), together with the molar mass and the bulk density. The optical constants δ and β are proportional to $(\rho / A) \lambda^2 f_1$ and $(\rho / A) \lambda^2 f_2$. The f_1 and f_2 cells, the molar mass and the density can be edited; **Save** overwrites the file. A value shown as *N/A* is not tabulated.

Note: f_1 , f_2 and the molar mass are per formula unit and enter the optical constants as a ratio. Change the mass only together with f_1 and f_2 , or every layer of that material changes its optical constants.

Benchmark

 **Calc > Benchmark** measures how reliably the optimizer converges, not how fast it runs:

- Every project (`.srcx`) in the `Benchmark` folder is reopened and fitted N times, with the fit settings stored in it.
- The final χ^2 of each run is tabulated with statistics, and the table is saved to `BenchResults\benchmark-<date>.dat`.
- A spread of final χ^2 values shows how often the swarm finds the same minimum; compare optimizer settings this way.

Configure the number of runs N (default 20) in **File > Settings > Calc & Fit**. For the time a fit takes, see the fitting time in the status bar.

See Also

- [Defining a Model](#)
- [File Formats > Henke Tables](#)

15 Application Settings

Paths and files, behavior, interface, calculation and fitting defaults, graphics

Access settings via **File > Settings**. The Settings dialog has five sections, navigated via the tree on the left.

Paths and Files

SETTING	DESCRIPTION
Default Project's Folder	Where new projects are saved by default
Work Folder	Base directory for all application data
Henke Library	Location of Henke atomic form factor tables
Jobs	Directory for batch job definitions
Benchmark output	Where benchmark results are saved
Benchmark	Location of benchmark model files
Fitting Output	Where fitting results are auto-saved

Register file associations (**xrcx**) — associates `.xrcx` files with X-Ray Calc 3 in Windows.

Behavior

SETTING	DESCRIPTION	DEFAULT
Automatically calculate when opening project	Calculate the active model when a project is opened	On
Live update structure panel during fitting	Show real-time parameter changes in the structure editor during fitting	Off
Automatically save results to output folder after fitting	Save the project as <name>-fitted.xrcx in the Fitting Output folder (or next to the project when that folder is not set) when a fit ends, and at each improvement during the fit if Live update is on	Off
Automatically check for updates	Check for new versions on launch	Off

Interface

Interface customization settings (reserved for future options).

Calc & Fit

SETTING	DESCRIPTION	DEFAULT
Number of CPU cores to use	Thread count for parallel computation	Auto (Use all)
Use the GPU for fitting	Evaluate the fitting swarm on the graphics card when one is usable (theta scans); see GPU Acceleration	On
Number of runs (Benchmark)	How many times each benchmark project is fitted (1–100)	20

Graphics

SETTING	DESCRIPTION	DEFAULT
Default line width	Width of chart series lines in pixels	2

See Also

- [Installation & First Launch](#)
- [File Formats > Settings File](#)

16 Universal Mirror Optimizer v3.7

Command-line design of multi-wavelength multilayer mirrors with LPSO: configuration, output, resume, figure of merit

The **Universal Mirror Optimizer** is a command-line tool that designs periodic multilayer mirrors (PMMs) optimized for *multiple characteristic wavelengths simultaneously*. Given a set of target elements (e.g. C K α and O K α), the optimizer searches for the best combination of materials, period thickness, layer ratio, number of periods, and interface roughness to maximize reflectivity across all targets at once.

This tool is part of the `xrccmd` command-line interface and uses the Henke optical constants database of the main X-Ray Calc 3 GUI application. Its reflectivity is a standalone single-precision port of the GUI's Parratt recursion with Névot–Croce roughness (the same 0.50299 σ^2 coefficient as the GUI engine); the substrate is a semi-infinite layer at bulk density.

Concept

Traditional multilayer mirror design targets a single wavelength. When an analytical instrument (e.g. an electron microprobe or X-ray fluorescence spectrometer) needs to measure multiple characteristic lines, the standard approach is to use separate mirrors for each element. The Universal Mirror Optimizer explores whether a single bilayer structure can provide useful reflectivity at two or more wavelengths simultaneously, even if it compromises peak performance at any single wavelength.

The optimizer uses the **Virtual Crystal Approximation** (VCA) to model mixed-material layers. Each layer's composition is defined as a weighted mixture of elements from a user-defined pool (e.g. W, Si, C). The atomic scattering factors f_1 , f_2 , the atomic mass and the bulk density of the constituents are each averaged with the layer fractions, and the dielectric constant is computed from these mixed values — not by averaging the constituents' dielectric constants.

Search Space

The optimizer searches over the following parameters:

PARAMETER	DESCRIPTION
Composition (per layer)	Element fractions for each layer in the bilayer period. Fractions are normalized to sum to 1.0.
d	Period thickness in Ångströms
γ (gamma)	Fraction of the period taken by layer 1, the upper layer of each period, whatever its material (0–1)
N	Number of periods (rounded to integer for evaluation)
σ (sigma)	Interface roughness in Ångströms. In bilayer mode it is shared by all interfaces, the substrate's included; in template mode the template's per-layer σ is used and this one applies to the substrate interface only
Density factor (per layer)	Multiplier on the bulk density of each layer. Each layer is built at this factor times its mixed bulk density (it had no effect before 3.9.3, when every layer was built at bulk density)

Optimization Algorithm

The optimizer uses its own implementation of **Lévy Flight Particle Swarm Optimization** (LFPSO), modelled on the GUI fitting engine (see [Chapter 9: Convergence Enhancements](#)) but not identical to it: $c_1 = c_2 = 2.05$ with the linear inertia $w_1 + w_2(1 - t/T)$ and no constriction factor, a Lévy scale falling from 0.1 to 0.01, and 30 % of the Lévy flights aimed at a random peer instead of the global best. Key features include:

- Adaptive PSO/Lévy flight switching: the probability of a Lévy step falls from 0.7 to 0.3 over the run, raised by 0.05 per stagnant iteration (at most +0.2, capped at 0.9)
- Reflective boundary handling with velocity damping
- Composition normalization (fractions always sum to 1.0)
- Window-based shake when the best FoM stalls: if it gains less than 1 % over `jamming_max` iterations, every particle but one is re-seeded around the global best
- Parallel fitness evaluation using all available CPU cores

The result reported at the end is the best solution found over the whole run, not the final position of the swarm.

CLI Usage

The Universal Mirror Optimizer is invoked via the `xrccmd` command-line tool with the `-u` switch:

```
xrccmd -u config.json [-l xrf_lines.json] [-v]
```

SWITCH	DESCRIPTION
<code>-u config.json</code>	Run the Universal Mirror Optimizer with the specified JSON configuration file
<code>-l xrf_lines.json</code>	XRF line table (shipped as <code>Shared\Universal\xrf_lines.json</code>). Required when a target is given by name or element range, or as an object without <code>lambda</code> ; without it such a run stops with “XRF lines not loaded”.
<code>-v</code>	Verbose mode — prints a line at every checkpoint save, and waits for <Enter> before exiting (do not use it in scripts). Shake events are not printed.

The `-pop` and `-iter` switches of the other `xrccmd` modes do not affect `-u`; the configuration file sets both.

Console Output

During optimization, the console displays a progress table updated every iteration:

```

Universal Mirror Optimizer v1.0
Lines: C(44.7A) O(23.6A)
Pool: W Si C
Structure: bilayer, d=[20..80], N=[20..100]
Mode: mixed compositions
Population: 100, Max iterations: 500
---
  Iter      FoM    R_C    R_O    Div  Best           Time
    0    0.0124  0.012  0.003  0.288  ...         0:01
    1    0.0298  0.031  0.009  0.271  ...         0:02
    ...
   500    0.1823  0.187  0.044  0.003  ...         3:12
---
Optimization complete.
Best FoM: 0.182300
Period d=54.94 A, gamma=0.697, N=36, sigma=0.72 A
Layer 1: C=100.0%
Layer 2: W=32.7% Si=35.8% C=31.5%

```

The banner also shows `Excluded:`, `Beam divergence:` and `Theta min:` lines when those options are set, and `Mode: pure elements (no mixing)` in pure-element mode. Columns: **Iter** = iteration number, **FoM** = figure of merit of the best solution so far (higher is better; it includes penalties, so it can be negative), **R_X** = peak reflectivity of that solution for each target, **Div** = normalised RMS spread of the swarm (about 0.3 for a random swarm, 0.5 at most; lower = more converged), **Best** = the layer pair and period of the best solution, **Time** = elapsed m:ss. A `Converged at iteration...` line appears if the stagnation limit stops the run, and `Package saved: ...` when the `.xrfx` package is written.

Configuration File

The optimizer is configured via a JSON file with the following sections. Every key shown is **required** unless it is listed as optional: a missing required key stops the run with an error. The optional keys and their defaults are `pure_elements` (false), `density_factor` (1.0), `delta_theta` (0), `theta_min` (0), `w_purity` (1.0),

`polarization` ("sp"), `scan_points` (200), `scan_half_range` (0 = adaptive), `n_ref` (50), `excluded_pairs` and `template_file` (none).

Targets

An array of target characteristic lines to optimize for, under the key `"lines"` or, for older files, `"targets"` (`"lines"` wins when both are present). An item is either an object, or a string: an element name such as `"O"` , or an element range such as `"B-Mg"` , whose wavelengths are looked up in the XRF line table (`-1`) with weight 1.

```
"targets": [
  {"element": "C", "lambda": 44.7, "weight": 1.0},
  {"element": "O", "lambda": 23.6, "weight": 1.0}
]
```

FIELD	TYPE	DESCRIPTION
<code>element</code>	string	Element name: labels the output, names the curve file and the <code>per_element</code> key, and finds the wavelength in the line table when <code>lambda</code> is omitted
<code>lambda</code>	number	Characteristic wavelength in Ångströms (e.g. C K α = 44.7 Å). Optional: taken from the line table (<code>-1</code>) when omitted
<code>weight</code>	number	Relative weight in the figure of merit (optional, default 1). Use higher values to prioritize specific targets.

Element Pool

The set of materials available for layer composition:

```
"element_pool": ["W", "Si", "C"]
```

Each element must have a corresponding Henke database file (`.bin`) in the Henke data directory. The optimizer can mix any combination of these elements in each layer.

Optional `"excluded_pairs": ["Mo/Si", ...]` forbids the listed layer pairs, in either order; it applies in pure-element mode only. Each name must be in the pool.

Structure

Defines the mirror topology and parameter search ranges:

```

"structure": {
  "type": "bilayer",
  "layers_per_period": 2,
  "pure_elements": true,
  "d": {"min": 20, "max": 80},
  "gamma": {"min": 0.2, "max": 0.8},
  "N": {"min": 20, "max": 100},
  "sigma": {"min": 0.5, "max": 3},
  "density_factor": {"min": 0.7, "max": 1.3}
}

```

FIELD	TYPE	DESCRIPTION
type	string	Required but not checked: the structure is always a two-layer period.
layers_per_period	integer	Required but not checked: the structure is always a two-layer period.
pure_elements	boolean	Optional, default <code>false</code> . When <code>true</code> , each layer is forced to use a single material from the pool (the dominant component). When <code>false</code> , layers can be arbitrary mixtures. Pure-element mode is recommended for most practical applications, and is required for <code>excluded_pairs</code> and templates.
d	range	Period thickness search range in Å. Ensure the range accommodates all target wavelengths ($d > \lambda/2$ for a valid Bragg condition).
gamma	range	Fraction of the period taken by layer 1, the upper layer of each period (0–1). Layer 1 thickness = $d \times \gamma$, Layer 2 = $d \times (1 - \gamma)$.
N	range	Number of bilayer periods. More periods give sharper peaks but tighter angular tolerance.
sigma	range	Interface roughness in Å: a range object, or a single number to fix the value. Shared by all interfaces in bilayer mode, the substrate's included.
density_factor	range	Optional: a range object, or a single number to fix the value (default 1.0 = bulk density). Each layer is built at the factor times its mixed bulk density, and <code>best_structure_xrc.json</code> carries the resulting density (before 3.9.3 the factor was searched but ignored).

Bragg condition: The period d must satisfy $\lambda/(2d) < 1$ for each target wavelength, otherwise no Bragg peak exists. For a target at $\lambda = 44.7$ Å, the minimum period is $d > 22.35$ Å. A target without a Bragg condition contributes zero to the FoM and is *not* penalised, so check the range yourself.

Fitness

Controls the figure of merit calculation:

```

"fitness": {
  "w_R": 1.0,
  "w_FWHM": 0.3,
  "R_min_threshold": 0.0001,
  "polarization": "sp",
  "delta_theta": 0.1,
  "theta_min": 3.0,
  "w_purity": 1.0
}

```

FIELD	TYPE	DESCRIPTION
<code>w_R</code>	number	Required. Weight for peak reflectivity in the FoM. Higher values emphasize maximizing R_{peak} .
<code>w_FWHM</code>	number	Required. Weight for the FWHM penalty. Higher values penalize broad peaks. Set to 0 to ignore bandwidth.
<code>R_min_threshold</code>	number	Required. Minimum acceptable peak reflectivity (after the divergence convolution). Each target below it subtracts 100 from the FoM to drive the swarm away from “dark” solutions.
<code>polarization</code>	string	<code>"s"</code> = s-polarisation; <code>"sp"</code> (the default, and any other value) = unpolarised average $(R_s + R_p)/2$.
<code>delta_theta</code>	number	Instrumental angular resolution in degrees. The calculated reflectivity curve is convolved with a Gaussian whose FWHM is <code>delta_theta</code> before extracting R_{peak} and FWHM. Default 0 = no convolution.
<code>theta_min</code>	number	Minimum acceptable Bragg angle in degrees (default 0 = none). A target whose Bragg angle falls below it subtracts 100 from the FoM and is left out of the FoM sum and of the purity. Prevents solutions at impractically low grazing angles.
<code>w_purity</code>	number	Weight for the spectral purity penalty (0–1). Controls how strongly higher-order Bragg peak contamination between targets is penalized. 0 = no purity penalty, 1 = full penalty. Default: 1.0. See Spectral Purity .
<code>scan_points</code>	integer	Minimum number of points in the scan around each Bragg peak; the grid is refined to at least 10 points per kinematic peak width, at most 20000. Default: 200.
<code>scan_half_range</code>	number	Fixed half-range of the scan in degrees. Default 0 = adaptive: $\pm \max(0.5^\circ, 4 \text{ kinematic widths})$ around both the kinematic Bragg angle and the refraction-corrected one ($\sin^2\theta' = \sin^2\theta_B + 2\delta$, δ averaged over the stack), never starting below $1.15 \theta_c$ or <code>theta_min</code> + 0.1° , and widened once (up to $\pm 5^\circ$) if the peak is clipped. The kinematic width uses the structure's own N.
<code>n_ref</code>	integer	Number of periods of the reference width FWHM_{ref} in the FoM (see Figure of Merit). Default: 50.

R_{peak} is the highest first-order local maximum inside the scan of the convolved curve, and the FWHM is measured between the linearly interpolated half-maximum crossings.

Optimizer

LFPSO algorithm parameters:

```
"optimizer": {
  "population": 100,
  "iterations": 500,
  "tolerance": 1e-8,
  "stagnation_limit": 50,
  "w1": 0.4,
  "w2": 0.5,
  "jamming_max": 5,
  "checkpoint_every": 50
}
```

FIELD	TYPE	EXAMPLE	DESCRIPTION
population	integer	100	Number of particles (candidate solutions) in the swarm. Larger populations explore more broadly but run slower per iteration.
iterations	integer	500	Maximum number of optimization iterations.
tolerance	number	1e-8	Required, but not used by the optimizer.
stagnation_limit	integer	50	Stop after this many consecutive iterations without improvement of the best solution. A shake resets that counter, so this takes effect only when jamming_max ≥ stagnation_limit ; with the example values the run always goes to iterations .
w1	number	0.4	PSO inertia weight (permanent velocity component).
w2	number	0.5	PSO inertia reducing factor (decays over iterations).
jamming_max	integer	5	Length of the stagnation window. At the end of each window, if the best FoM improved by less than 1 %, the swarm is shaken: all particles but one are re-seeded within ±10 % of each range around the global best (in pure mode each layer also has a 50 % chance of a random element), and their velocities are reset.
checkpoint_every	integer	50	Save a checkpoint file every N iterations. 0 turns periodic checkpoints off (before 3.9.3 it stopped the run with a division-by-zero error); a final checkpoint is always written.

Recommended settings: For a 3-element pool with 2 targets, population: 100 and iterations: 500 is a good starting point. Increase population for larger element pools. Reduce iterations for quick exploratory runs.

Top-Level Fields

```
"substrate": "Si",
"henke_path": null,
"output_dir": "./results",
"resume_from": null
```

FIELD	TYPE	DESCRIPTION
substrate	string	Required. Substrate material: a semi-infinite layer at bulk density under the mirror.
henke_path	string or null	Required (<code>null</code> allowed). Path to a directory named <code>Henke</code> : only the parent of this path is used, and the files are read from its <code>Henke</code> subfolder, so the folder must have that name. <code>null</code> uses <code>Henke\</code> next to the executable.
output_dir	string	Required. Directory where all output files are written. Created automatically if it does not exist.
resume_from	string or null	Required (<code>null</code> allowed). Path to a checkpoint file to resume from. <code>null</code> starts a fresh optimization. See Checkpoint & Resume .
template_file	string or null	Optional. A template file of period definitions keyed by layer pair, such as <code>"Mo/Si"</code> (see below).

Period Templates

With `template_file` set, in pure-element mode, a period whose layer-1/layer-2 pair matches a template key is built from the template instead of as a plain bilayer. Each template layer gives a `material` , a `thickness` (`"gamma"` , `"1-gamma"` or a fixed value in Å), a `sigma` and a `density` ; fixed interlayers are subtracted from the nearest main layer, and a period whose interlayers would need a negative main-layer thickness is penalised by 1. Optional `caps` / `cap` add a capping layer whose thickness and variant are searched. Template materials are loaded in addition to the pool.

Output Files

The optimizer writes several files to the `output_dir` upon completion:

Note: a fresh run (`resume_from` `null`) first deletes `best_curves/` , `best_structure.json` , `best_structure_xrc.json` , `population.json` , `progress.log` and `checkpoint.json` in `output_dir` . Use a new directory to keep an earlier result.

FILE	DESCRIPTION
<code>best_structure.json</code>	The optimized design: FoM, composition, period, gamma, N, sigma, and per-target R_{peak} and FWHM, plus a summary structure. Not importable into the GUI.
<code>best_structure_xrc.json</code>	The same design as an X-Ray Calc 3 structure (<code>Stacks</code> / <code>Subs</code>): the file to open with the GUI's Import structure...
<code>best_curves/Element.dat</code>	Tab-separated reflectivity curves for each target element, with a header line. Two columns: angle (degrees) and reflectivity, convolved with <code>delta_theta</code> . One file per target.
<code>population.json</code>	The 10 best particles' personal bests, ranked by FoM (FoM, composition, d, γ , N, σ). They are not deduplicated: late in a run they can be near-identical.
<code>checkpoint.json</code>	Full optimizer state (all particle positions, velocities, and personal bests). Used for resuming interrupted runs.
<code>progress.log</code>	Plain text log of every iteration's FoM, per-target R_{peak} , and population diversity, from iteration 0. Matches the console output. Rewritten from scratch on resume.
<code>config.json</code> , <code>manifest.json</code>	A copy of the configuration and a manifest of the results, for the package.
<code>config.xrfx</code>	Written next to the configuration file: a zip of <code>output_dir</code> without the checkpoint.

best_structure.json Format

The output structure file contains two sections:

- **optimizer_result** — numerical results: FoM, structural parameters (d, γ , N, σ), layer compositions, and per-element R_{peak} and FWHM .
- **structure** — a summary of the stack: the periodic bilayer with layer thickness and roughness (the third value is always 1.0), and the substrate written as `[0, 1, 8.0]` . For the GUI use `best_structure_xrc.json` .

Reflectivity Curve Files

Each `.dat` file in the `best_curves/` subdirectory contains the reflectivity for that target wavelength over the same window the figure of merit used (see `scan_half_range`), without the re-scan of a clipped peak; its length varies, and the convolution trims points at each end. The data is tab-separated:

```

Theta(deg)      Reflectivity
3.2400  1.23456789e-03
3.2900  2.34567890e-03
...
```

Checkpoint & Resume

Long optimization runs can be interrupted (Ctrl+C) and resumed later. The optimizer handles this gracefully:

1. **Automatic checkpoints** — saved every `checkpoint_every` iterations to `checkpoint.json` in the output directory.
2. **Ctrl+C handling** — Ctrl+C lets the current iteration finish, then the optimizer writes all the normal result files, the `.xrfx` package and a final checkpoint.
3. **Resume** — set `"resume_from"` in the configuration file to the path of a checkpoint file. The optimizer restores all particle positions, velocities, personal bests, and the global/all-time best. Every final checkpoint is stamped with the configured `iterations`, even after Ctrl+C or an early stop, and the run resumes after that stamp: to continue you must **raise** `iterations`, otherwise nothing more is run.

Example resume configuration:

```
{
  "targets": [...],
  "element_pool": ["W", "Si", "C"],
  ...
  "optimizer": {
    "population": 100,
    "iterations": 1000,
    ...
  },
  "output_dir": "./results",
  "resume_from": "./results/checkpoint.json"
}
```

Important: When resuming, the swarm size comes from the checkpoint (the `population` value is ignored), and the `element_pool` must match it. `iterations` must be larger than the checkpoint's stamp. `progress.log` restarts.

Figure of Merit

The figure of merit (FoM) combines peak reflectivity and angular bandwidth for all target wavelengths:

$$\text{FoM} = \sum_i [\text{weight}_i \times (w_R \times R_{\text{eff},i} - w_{\text{FWHM}} \times \text{FWHM}_i / \text{FWHM}_{\text{ref},i})]$$

where:

- **$R_{\text{eff},i}$** — effective reflectivity for target i , which equals $R_{\text{peak},i}$ when `w_purity` is 0, or purity-adjusted reflectivity when `w_purity` > 0 (see [Spectral Purity](#))
- **$R_{\text{peak},i}$** — peak reflectivity (0–1) for target i , the first-order maximum of the convolved curve in the scan window (see `scan_half_range`)
- **FWHM_i** — angular full width at half maximum (degrees) of the Bragg peak

- **FWHM_{ref,i}** — reference width $\lambda_i / (N_{\text{ref}} \cdot d \cdot \cos \theta_{B,i})$, converted to degrees, with $\theta_B = \text{asin}(\lambda/2d)$ and $N_{\text{ref}} = n_{\text{ref}}$ (default 50), *not* the structure's N: a reference tied to N would penalise adding periods, which only raises the reflectivity. It normalizes the FWHM penalty so it is comparable across targets.
- **weight_i** — per-target weight from the configuration
- **w_R, w_{FWHM}** — global weights for reflectivity and bandwidth terms

Higher FoM is better. Each target with R_{peak} below `R_min_threshold`, or with θ_B below `theta_min`, subtracts 100 from the FoM to steer the swarm toward functional solutions; the reported FoM includes these penalties, so it can be negative. A target without a Bragg condition ($\lambda \geq 2d$) contributes zero and is not penalised.

Spectral Purity

When optimizing a mirror for multiple characteristic wavelengths, higher-order Bragg reflections can cause **cross-contamination** between measurement channels. For example, a mirror tuned to O K α (23.6 Å) at its first-order Bragg angle will also reflect wavelengths at $\lambda/2 = 11.8$ Å via the second order — which nearly coincides with Na K α (11.9 Å). When both O and Na are among the target elements, any Na fluorescence reaching the detector at the O measurement angle is misattributed to O.

The spectral purity penalty addresses this by computing a **cross-reflectivity matrix** and penalizing solutions where contamination is significant.

Cross-Reflectivity Matrix

For each pair of targets i and j , the optimizer computes:

$$\text{CrossR}[j, i] = R(\theta_j, \lambda_i)$$

This is the reflectivity of light with wavelength λ_i hitting the mirror at the Bragg angle of target j . A high value of $\text{CrossR}[j, i]$ means that when the spectrometer measures target j , it also picks up signal from target i 's wavelength via higher-order reflection. The calculation uses the full multilayer reflectivity at each (θ, λ) point — all Bragg orders are naturally included. CrossR is a single-point value at the kinematic Bragg angle of target j , $\theta_j = \text{asin}(\lambda_j/2d)$, without refraction correction or divergence convolution; it is computed only when `w_purity` > 0 and only between targets that have a valid Bragg angle, and every line counts equally regardless of its source intensity.

Purity and Effective Reflectivity

For each target j , the **purity** is defined as:

$$\text{Purity}_j = R_{\text{peak},j} / (R_{\text{peak},j} + \sum_{i \neq j} \text{CrossR}[j, i])$$

Purity ranges from 0 (entirely contaminated) to 1 (no contamination); it is set to 0 when $R_{\text{peak},j}$ and the contamination sum are both 0. The effective reflectivity used in the FoM is then:

$$R_{\text{eff},j} = R_{\text{peak},j} \times (1 + w_{\text{purity}} \times (\text{Purity}_j - 1))$$

When `w_purity = 0`, $R_{\text{eff}} = R_{\text{peak}}$ (no penalty). When `w_purity = 1`, $R_{\text{eff}} = R_{\text{peak}} \times \text{Purity}$, giving maximum penalization of contaminated channels. Intermediate values blend between the two.

Physical Background

Higher-order contamination depends on two structural factors:

- **γ (layer ratio)** — controls which orders are suppressed. $\gamma = 0.5$ suppresses the even-order reflections (2nd, 4th, ...); they vanish exactly only in the kinematic limit with ideal, non-absorbing interfaces, and in the full dynamical calculation they are strongly reduced rather than zero. Values far from 0.5 produce strong even-order peaks. Second-order contamination is typically the most significant.
- **σ (roughness)** — interfacial roughness attenuates order m by roughly $\exp[-(2\pi m\sigma/d)^2]$, a Debye–Waller-like factor that is Gaussian in m , so high orders die quickly once σ is a sizeable fraction of d/m . Orders above 4–5 are generally negligible for typical roughness values.

Practical note: The overlap pattern is inherent to the target wavelength set — second-order overlap at $\lambda/2$ is independent of the period d . The optimizer can only mitigate overlaps through γ (order suppression), material choice (absorption edges), and roughness. For large element pools with many close-spaced wavelengths, use `w_purity = 1.0` and a population ≥ 500 to give the optimizer enough room to find purity-optimized solutions.

Example Workflow

This example designs a multilayer mirror for simultaneous C K α (44.7 Å) and O K α (23.6 Å) characterization using W/Si/C materials on a Si substrate.

Step 1: Create the configuration file

Save the following as `dual_mirror.json`:

```
{
  "targets": [
    {"element": "C", "lambda": 44.7, "weight": 1.0},
    {"element": "O", "lambda": 23.6, "weight": 1.0}
  ],
  "element_pool": ["W", "Si", "C"],
  "structure": {
    "type": "bilayer",
    "layers_per_period": 2,
    "d": {"min": 20, "max": 80},
    "gamma": {"min": 0.2, "max": 0.8},
    "N": {"min": 20, "max": 100},
    "sigma": {"min": 0.5, "max": 3},
    "density_factor": {"min": 0.7, "max": 1.3}
  },
  "fitness": {
    "w_R": 1.0,
    "w_FWHM": 0.3,
    "R_min_threshold": 0.0001,
    "polarization": "sp"
  },
  "optimizer": {
    "population": 100,
    "iterations": 500,
    "tolerance": 1e-8,
    "stagnation_limit": 50,
    "w1": 0.4,
    "w2": 0.5,
    "jamming_max": 5,
    "checkpoint_every": 50
  },
  "substrate": "Si",
  "henke_path": null,
  "output_dir": "./dual_mirror_results",
  "resume_from": null
}
```

Step 2: Run the optimizer

```
xrccmd -u dual_mirror.json -v
```

The `-v` flag enables verbose output, showing checkpoint saves, and waits for <Enter> at the end. A typical run with population 100 and 500 iterations takes a few minutes on a modern multi-core CPU.

Step 3: Review results

After completion, examine the output directory:

- `dual_mirror_results/best_structure.json` — the optimal mirror design and its performance
- `dual_mirror_results/best_curves/C.dat` — reflectivity curve at C K α (44.7 Å)

- `dual_mirror_results/best_curves/0.dat` — reflectivity curve at O K α (23.6 Å)
- `dual_mirror_results/population.json` — top 10 alternative solutions
- `dual_mirror_results/best_structure_xrc.json` — the design to open with the GUI's **Import structure...**

The reflectivity curve files can be plotted in any graphing tool (Origin, gnuplot, Python/matplotlib) or imported into the X-Ray Calc 3 GUI for comparison with experimental data.

Step 4: Continue optimization (optional)

If you want to refine the result further, increase the iteration count and add a resume path:

```
{
  ...
  "optimizer": {
    "population": 100,
    "iterations": 1000,
    ...
  },
  "resume_from": "./dual_mirror_results/checkpoint.json"
}
```

Then re-run: `xrccmd -u dual_mirror.json -v`

Tips and Best Practices

- **Start small, then refine.** Begin with a small population (20–50) and few iterations (50–100) to verify the configuration produces sensible results, then increase for production runs.
- **Element pool size.** Each additional element adds composition dimensions for each layer. Pools of 3–5 elements work well. Very large pools may require larger populations.
- **Period range.** Ensure $d_{\min} > \lambda_{\max}/2$ so all targets have valid Bragg conditions. A generous range (e.g. 20–80 Å) allows the optimizer to find the best compromise period.
- **Weighting targets.** If one target is more important, increase its `weight` in the configuration. The optimizer will prioritize that target's reflectivity at the expense of the others.
- **Interpreting diversity.** A random swarm starts near 0.3. A diversity value near zero means all particles have converged to similar solutions. If diversity drops quickly but FoM is still low, consider increasing the population or widening the search ranges.
- **Population JSON.** The `population.json` file contains the 10 best personal bests; they are not deduplicated, so late in a run several may be near-identical. Early or diverse runs often reveal physically different mirror designs (different compositions or periods) with similar overall FoM. Examining them can provide insight into the design trade-space.

See Also

- [MCP Server > Optimisation and Jobs](#)
- [Fitting \(Optimization\)](#)

17 MCP Server (XRC_MCP)

Connect an AI assistant to the calculation engine through the Model Context Protocol

Overview

XRC_MCP is a console program that exposes the X-Ray Calc 3 calculation engine to AI assistants over the **Model Context Protocol** (MCP). An MCP client such as Claude Code starts `XRC_MCP.exe`, talks to it as JSON-RPC 2.0 over standard input and output, and calls its **tools**: calculate a reflectivity curve, score a mirror design, run the universal optimizer, fit a measured curve, and read or write `.xrcx` project files.

The engines are the GUI's own. The curve, the χ^2 and the figure of merit the server returns are the numbers X-Ray Calc 3 and XRFCalc display for the same structure, and a fit written by the server opens in the GUI as an ordinary project file. This lets you have an assistant prepare, run and compare fits while you verify every result in the GUI.

PROPERTY	VALUE
Executable	<code>XRC_MCP.exe</code> , 64-bit only, in the same folder as the GUI
Transport	JSON-RPC 2.0 over stdio; one JSON object per line, UTF-8
Tools	18, grouped below
Long jobs	<code>optimize_mirror</code> and <code>fit_xrr</code> run in the background and are polled
Record	Every call and result appended to <code>log\calls.jsonl</code>

Installation and Registration

The server needs no configuration file. It takes one mandatory argument, the **working directory** that becomes its sandbox (see [below](#)):

```
XRC_MCP.exe --workdir "D:\Experiments\RuC-2026-09"
```

Register it with Claude Code once per experiment directory. The name `xrc` is how the assistant refers to the server:

```
claude mcp add -s user xrc -- "C:\Program Files\X-RayCalc3\XRC_MCP.exe" --workdir "D:\Experiments\RuC-
```

Other MCP clients register a stdio server the same way: the command is the path to `XRC_MCP.exe`, the arguments are `--workdir` and an absolute path. Nothing in the server depends on the current directory of the process.

To verify the registration, ask the assistant to call `describe_server`. The reply states the server version, the git revision it was built from, the Henke table directory in use, the unit conventions, and a one-line summary of

every tool.

OmniThreadLibrary 3.08: If you build the server yourself, it needs OmniThreadLibrary 3.08 or later. With an older library a job stays at iteration 0 for ever and the server will not exit. See [Source Code](#) > [Dependencies](#).

The Working Directory

Every path the server reads or writes lies under the working directory. It is created on first start with four subfolders:

```
<workdir>\
  projects\      .xrcx files the server writes (save_project) and reads (load_project)
  jobs\          one folder per job: request, progress log, curves, results
  inbox\         measured curves that YOU put there; the server never writes here
    <specimen>\  one folder per specimen, any number of data files inside
  log\
    calls.jsonl  every tools/call and its result, appended, never truncated
```

A path argument that resolves outside the working directory (a rooted path, a drive letter, a UNC path, or a `..` segment) is refused with the error `path_outside_workdir`; it is never silently remapped.

The journal is the record. `log\calls.jsonl` holds one line per call with a timestamp, the tool name, the full arguments and the full result or error. Long numeric arrays such as curves are replaced by their length and the SHA-256 of the file they were written to. Keep it with the experiment.

Units and the Structure JSON

The conventions are the program's: thickness, roughness σ and wavelength in **ångström**, density in **g/cm³**, angles as the grazing angle **θ in degrees** (not 2θ). Wherever a wavelength is expected, a photon energy in eV is accepted instead and converted with $\lambda[\text{\AA}] = 12398.42 / E[\text{eV}]$; the wavelength actually used is echoed back. (The engine itself looks the Henke tables up with 12398.6; `describe_server` reports both constants.)

Every tool that takes or returns a layered structure uses one JSON form, listed from the substrate to the surface:

```
{
  "substrate": {"material": "SiO2", "density": 2.2, "sigma": 3.0},
  "buffer":    {"material": "Ru", "thickness": 197.0, "sigma": 3.0},
  "stacks": [
    {"N": 30, "layers": [
      {"material": "Ru", "thickness": 14.7, "sigma": 3.0, "density": 12.4},
      {"material": "C", "thickness": 53.8, "sigma": 3.0, "density": 2.2}
    ]}
  ],
  "cap":       {"material": "Ru", "thickness": 20.0, "sigma": 3.0}
}
```

- `material` is a Henke table name, matched case-insensitively (`list_materials` gives the valid names).
- `density` is optional and defaults to the bulk value from the table; the value used is echoed. `thickness` is required; `sigma` defaults to 0 and `N` to 1.
- `buffer` (between substrate and first stack) and `cap` (on top) are optional. Several stacks are allowed.
- Numbers in the JSON replies are rounded to 6 significant digits; the curve files keep full precision (θ to 7 digits, R as `%.8e`).

Reference Tools

TOOL	DESCRIPTION
<code>describe_server</code>	Call it first. Reports the server version and git revision, the Henke directory, the unit conventions, the material-name syntax, the structure form, the limits, the working-directory layout, and a summary of every tool.
<code>list_materials</code>	The Henke tables the server can use, optionally filtered by an element pool. Each entry gives the table name, formula, bulk density and atomic mass.
<code>optical_constants</code>	δ and β of $n = 1 - \delta + i\beta$ for one material at one wavelength or energy, plus n , k and the nearby absorption edges. Unknown materials are refused with <code>unknown_material</code> .
<code>list_templates</code>	The multilayer design templates (one period each: sub-layers with thickness rule, σ , density, capping variants) that <code>optimize_mirror</code> can start from, optionally filtered by a material pool.

Calculation Tools

TOOL	DESCRIPTION
<code>calc_reflectivity</code>	Specular reflectivity of a structure over a θ range at one wavelength: <code>theta_min</code> (default 0.05°), <code>theta_max</code> (default 5°, at most 90°), <code>points</code> (default 2000, at most 100 000), <code>polarization</code> (below) and an optional beam divergence <code>delta_theta</code> (default 0; when set, the FWHM in degrees of the Gaussian the curve is convolved with — a width the $\pm 0.1^\circ$ kernel cannot represent on the grid is refused with <code>invalid_argument</code>). The calculated R is clamped from below to <code>r_min</code> (default 10^{-7}). Returns the curve inline when short enough and always as <code>jobs\ <id>\curve.dat ; period_A , n_periods</code> and <code>structure_used</code> (densities filled in); the Bragg peaks as <code>{order, theta_deg, r_peak, fwhm_deg} ;</code> and <code>critical_angle_deg</code> . <code>theta_deg</code> is the calculated maximum, refraction shift included, not the kinematic θ_B : each order of the period of the first repeating stack (counted from the surface) is looked for between $\theta_B - 0.06^\circ$ and the refraction-corrected angle $+ 0.06^\circ$, must lie more than 0.05° above θ_c and stand more than three times above the window minimum, and is left out otherwise. FWHM is taken between the interpolated half-maximum crossings, 0 when the curve does not fall to half on both sides. <code>critical_angle_deg</code> is estimated as $\sqrt{2(\delta)}$ from the thickness-weighted δ of the top 500 Å (substrate included where the film is thinner), not read off the curve. This is the GUI's curve for the same structure (see Calculation).
<code>polarization</code>	In every tool that takes it: <code>"sp"</code> (default, the unpolarized average $(R_s + R_p)/2$) or <code>"s"</code> . The engine has no pure-p path, so <code>"p"</code> is computed as <code>"sp"</code> and echoed as such.
<code>evaluate_lines</code>	Scores a structure against a set of X-ray emission lines (<code>name</code> , <code>lambda</code> or <code>energy</code> , <code>weight</code>) with the universal optimizer's own figure of merit; optional <code>fitness</code> overrides. A line may also be a bare symbol or a range such as <code>"B-Si"</code> ; at most 16 lines, and exactly one stack must repeat. Refused with <code>server_busy</code> while <code>optimize_mirror</code> runs. The number is directly comparable with what <code>optimize_mirror</code> reports (see Universal Mirror > Figure of Merit).

Optimisation and Jobs

Two tools take longer than a request should wait for. They return a `job_id` at once and run in a background thread; the client waits with `job_wait` or polls `job_status`. One job runs at a time, and up to 16 more can queue behind it.

TOOL	DESCRIPTION
<code>optimize_mirror</code>	Searches for the periodic multilayer that best serves a set of emission lines with the XRFCalc particle swarm. Takes the optimizer configuration (lines, <code>element_pool</code> , excluded pairs, structure ranges, fitness, optimizer settings, substrate, template), a <code>seed</code> and <code>top_k</code> (default 5). The result holds the <code>top_k</code> distinct structures with genome, figure of merit, per-line results and the <code>.xrfx</code> package written for each.
<code>job_status</code>	State of a job without waiting: <code>queued</code> , <code>running</code> , <code>finished</code> , <code>failed</code> , <code>cancelled</code> or <code>unknown</code> , with the current iteration and budget, the best value so far, elapsed seconds and the last message.
<code>job_wait</code>	Waits for a job and returns the <code>job_status</code> object plus <code>waited_s</code> . It returns as soon as the job is <code>finished</code> , <code>failed</code> or <code>cancelled</code> , or after <code>wait_s</code> seconds (default 300, at most 900). While it waits the server answers nothing else, so <code>cancel_job</code> cannot reach it; keep <code>wait_s</code> below the client's tool timeout.
<code>job_result</code>	The result of a finished job. Fails with <code>job_not_finished</code> while it runs, <code>job_failed</code> when it ended in an error, <code>job_cancelled</code> when it was stopped.
<code>cancel_job</code>	Asks a job to stop and returns its state. A running job usually stops within a second or two; poll <code>job_status</code> to see it become <code>cancelled</code> .

Determinism and lifetime: With the same `seed` and inputs, `optimize_mirror` and `fit_xrr` give identical results on the same device; the seed used is echoed. With `device: "auto"` a machine with a GPU and one without can differ in the last digits, so pin `"cpu"` or `"gpu"` for bit-for-bit repeats. Jobs do not survive the server: if the process ends, running jobs are lost and `job_status` reports `unknown` after a restart. Each job's folder `jobs\<job_id>\` keeps its request, progress log and outputs.

Fitting

TOOL	DESCRIPTION
<code>fit_xrr</code>	Fits a layer model to a measured curve with the GUI's LFPSO, minimising the same χ^2 X-Ray Calc 3 displays (see Fitting). Returns a <code>job_id</code> ; the result is read with <code>job_result</code> .

The main arguments:

ARGUMENT	MEANING
measurement_id or curve	The measured data: an inbox id such as "RuC-07/xrr.dat", or the curve inline. Every measured intensity must be positive, since the χ^2 takes its logarithm.
structure	The start model in the structure JSON.
free	Which values may vary. Each entry addresses one layer by its stack index (from the substrate up, or "cap" / "buffer") and layer index, and lists the parameters that take part: "thickness", "sigma", "density". By default a repeating stack keeps the start model's period, so freeing its thicknesses fits only their ratio. Add {"target":"period","stack":k} to fit the period d of repeating stack k as well; at least one thickness of that stack must also be free, and it cannot be combined with profile. The substrate, scale, background and resolution are not free parameters; asking for them is answered with not_fittable (the scale is solved inside the objective instead, see scale_solve below). period_mode in the result says how each stack was treated.
bounds	One min / max pair per free parameter (stack , layer , parameter), or per free period (target: "period"). Omitted bounds default to the start value $\pm 30\%$ (a period too); the start value must lie inside the range. A parameter that starts at 0 (for example an omitted sigma) has no default range and needs explicit bounds; an omitted density is first replaced by the bulk value. Sigma and density bounds are clamped at 0; thickness and period bounds must be above 0. Every fitted value stays inside its bounds, and the result's bounds_used lists the ranges that applied.
lambda or energy , theta_range	The wavelength (or photon energy) of the measurement: required with an inline curve ; with measurement_id it defaults to the .xrdml file's own wavelength (the weighted K α doublet, 1.541874 Å for Cu) or else meta.json's lambda , and an explicit value overrides both. theta_range is an object {"min", "max"} in degrees θ (never 2θ); default the whole curve.
scale , scale_auto	The anchored multiplier applied to the measured intensities before the fit (the manual's Normalize step), default 1, stored with the curve in measured.dat and fit.xrcx . "scale_auto": true (or "scale": "auto") is the GUI's Normalize (Auto) : the largest measured intensity below auto_theta_max (default 0.5°) is set equal to the start model's reflectivity at that angle.
scale_solve , scale_solve_window	Default true : the measured scale is treated as a nuisance parameter and solved in closed form for every candidate structure, held between $1/(1 + \text{scale_solve_window})$ and $(1 + \text{scale_solve_window})$ times the anchored scale (default 0.2, a fraction: a factor of 1.2 either way), so the χ^2 is the best any scale in that window can give. The result reports chi2_scale ("solved" or "anchored"), scale_ratio (the solved factor over the anchor), scale_solved , scale_clamped and chi2_scale_definition ; the files and the .xrcx keep the anchored scale, and residual.dat says in its header which scale it is at. "scale_solve_window": 0 pins the solved scale to the anchor. With "scale_solve": false the χ^2 is the anchored one of every earlier version. The GUI's Solve scale in χ^2 and Window are the same option with the same defaults; the physics is in Solving the Measured Scale .

<code>smooth</code>	<code>{"passes": n}</code> , 0 to 10, runs the GUI's Data > Smooth on the measured curve before the fit: each pass is a moving average of window 5 on the linear intensities, which leaves the first points as they are and flattens the last few, so trim the ends with <code>theta_range</code> . Order: scale, then smooth, then the <code>theta_range</code> trim, as in Data Conditioning .
<code>optimizer</code>	<code>population</code> (default 500), <code>iterations</code> (default 100), <code>tolerance</code> (default 0.005: the run stops early once χ^2 falls below it), <code>shake</code> , <code>range_seed</code> and <code>use_constriction</code> (all default true), and <code>device</code> : "auto" (default, the GPU when <code>describe_server</code> names one in <code>server.gpu</code>), "cpu" or "gpu" (refused without a usable GPU). The result carries <code>optimizer_used</code> with every default filled in, and <code>device_used</code> . The reported χ^2 is always recomputed on the CPU (<code>chi2_recalc</code> , with <code>consistency_warning</code> when the two disagree); <code>gpu_error</code> says why a run asked to use a GPU fell back to the CPU. See GPU Acceleration .
<code>profile</code> , <code>paired</code> , <code>seed</code>	<code>profile</code> fits a polynomial (order <code>optimizer.poly_order</code> , default 1) of each free parameter — thickness, sigma and density alike — over the periods of the one repeating stack (the GUI's polynomial mode; exactly one stack must have $N > 1$). <code>paired</code> (for example <code>["sigma", "density"]</code>) keeps chosen parameters at one value per layer. The per-period arrays start at the period nearest the surface. <code>seed</code> is the random seed.
<code>polarization</code> , <code>resolution</code> , <code>chi2</code> , <code>r_min</code>	<code>polarization</code> as in <code>calc_reflectivity</code> (default "sp"). <code>resolution</code> : FWHM in degrees θ of the Gaussian the calculated curve is convolved with (default 0.015, held fixed); the kernel stops at $\pm 0.1^\circ$ and is normalised, χ^2 leaves out the points within about 0.1° of each end of the fitted range, and a grid too coarse or too short for the kernel is refused. <code>chi2</code> : <code>theta_weight</code> 0–5 (default 0), <code>point_weight</code> (default true: a point more than $3\times$ its moving average is weighted by that ratio) and <code>movavg_window</code> (default 0.05 of the points); the sum is $1000/(n-1) \sum [(\log I - \log R)/\log R]^2 \cdot w$ (see the cost function). <code>r_min</code> (default 10^{-7}): the floor the calculated reflectivity is clamped to; no measured point is dropped. It is also the chart minimum stored in the <code>.xrcx</code> .

The result holds the fitted structure (with the expanded per-period profile when `profile` was set), χ^2 beside `chi2_start` and `chi2_plain` (the same sum without point and θ weights, at the scale solved for `chi2` ; reported, never minimised; `chi2_start` is scored at the start model's own solved scale, `scale_start_ratio`), the scale mode and values, `period_mode` , `out_of_bounds` , `iterations_run` , `elapsed_s` , `background` (always 0), the start model, a `report` (the Bragg orders measured against calculated, the plateau edge, the Kiessig fringe contrast, the residual by band, values within 5 % of the range from a bound; also written as `report.json` with the scale mode beside every χ^2 it holds; the report compares the model with the *anchored* measured curve, so with `scale_solve` its ratios and bands differ from `residual.dat` by $\log_{10}(\text{scale_ratio})$), the measured, calculated and residual curves as files under the job folder, and the path of the `fit.xrcx` written for the fit. Open that file in the GUI to check the fit yourself.

Inbox: Measured Curves

Measured curves reach the server through the `inbox\` folder, one subfolder per specimen. You copy the files there; the server only reads them and never modifies them.

TOOL	DESCRIPTION
<code>list_measurements</code>	Every file in the inbox, optionally filtered by specimen, with its id <code>"<specimen>/<file>"</code> , size, SHA-256, modified time, and the specimen's metadata file if present.
<code>get_measurement</code>	One curve as $[\theta, \text{intensity}]$ pairs, parsed the way the GUI parses a data file (see File Formats): two columns separated by tab or space, comma accepted as the decimal mark, non-numeric lines reported as the header. An <code>.xrdm1</code> file is read raw, as Data > Load reads it: its 2θ axis is brought to θ , its counts to counts per second normalised to 1 at the maximum, and the wavelength the file implies overrides <code>meta.json</code> (<code>lambda_source</code> says which). A non-positive intensity is replaced by the smallest positive one before it, and <code>"theta_unit": "2theta"</code> in <code>meta.json</code> halves the angles of a two-column file. <code>max_points</code> (default 2000) decimates a long curve for inline return; the file stays raw.

assess_xrr: is this curve worth fitting?

`assess_xrr` runs the eight measurement checks of the GUI's [Data > Assess XRR quality](#) on an inbox file, before anyone fits it. It reads the `.xrdm1` itself, because the decisive checks are count rates and the counting time exists only there; a two-column file still works, and every check that needs a fact the file does not carry answers `"unknown"`. Nothing is written anywhere.

ARGUMENT	MEANING
<code>measurement_id</code>	The inbox file, as <code>list_measurements</code> reports it.
<code>structure</code>	The design, optional: the same structure object <code>calc_reflectivity</code> takes. Unlike the other tools this one accepts a bare substrate, <code>"stacks": []</code> , for a glass reference. With a design the checks that need a period, a total thickness or a critical angle can answer, and the design's reflectivity on the measured range, convolved with <code>resolution</code> (default <code>fit_xrr</code> 's 0.015°), says what the measurement could have shown.
<code>lambda</code> or <code>energy</code>	For the design; defaults to the file's wavelength or <code>meta.json</code> 's.
<code>detector_max_cps</code>	The detector's linear count-rate limit, judged against the rate the detector itself saw (attenuation factors out). Without it the counting check reports the peak rate under <code>"unknown"</code> ; no limit is built in.
<code>sample_length_mm</code> , <code>beam_width_mm</code>	Place the footprint knee $\text{asin}(\text{beam} / \text{length})$.
<code>order_visible_factor</code> , <code>min_points_per_fringe</code>	An order counts as visible this many times above the background (default 3); the sampling check warns below this many points per Kiessig fringe (default 3) and fails below 2 whatever is given.

The result carries `checks` — `counting`, `plateau_vs_first_order`, `total_reflection`, `orders_visible`, `range_below_background`, `sampling`, `footprint`, `zeros`, each {`value`, `threshold`, `verdict`, `why`} with the numbers behind it — the background, the design's period, thickness and critical angle, the overall `verdict` (the worst check) and `summary_text`, one line per check for the client to file with the specimen. Every angle is `theta_deg` with the file's `two_theta_deg` beside it. No count-rate or fraction limit is built in. The fixed rules are: a first order above the plateau fails; a scan that starts at its own maximum, or at or above the design's critical angle, fails; fewer than two points per fringe fails. The warnings compare the measurement with the design scaled to the measured plateau (a plateau-to-first-order ratio above the design's; fewer visible orders than it predicts), with the footprint knee (the low-angle maximum below it), or flag any zero or negative count. `range_below_background` only reports the fraction and is always "unknown". The background is the median of the last 100 points; the fringe spacing is $\lambda/(2 \times \text{total thickness})$ and the order spacing $\lambda/(2d)$, converted to degrees; the design is always computed unpolarized (`sp`).

Projects

TOOL	DESCRIPTION
<code>save_project</code>	Writes <code>projects\<name>.xrcx</code> that opens in the GUI with no conversion: the structure as the active model, the calculated curve beside it (without a job: $\lambda = 1.54043 \text{ \AA}$ unless given, θ 0.05–5°, 2000 points, <code>s</code> polarization as the GUI's <code>params.dsc</code> has it — so it can differ from a default <code>calc_reflectivity</code> curve, which is <code>sp</code>), and optionally a measured curve (an inbox id or a job's curves via <code>curves.job_id</code>). A project saved from a fit job carries the job's own fit settings, so it opens as the fit that ran. Refuses to overwrite unless <code>overwrite: true</code> .
<code>load_project</code>	Reads a project back: the active model's structure in the structure JSON, its profile extensions, the calculation parameters (angles always as θ), and a summary of the embedded curves.
<code>list_projects</code>	The <code>.xrcx</code> files in <code>projects\</code> with size, SHA-256 and modified time. Nothing is written.

Example Session

A typical exchange, with the assistant's tool calls in order. The curve `RuC-07\xrr.dat` was copied into the inbox beforehand.

1. `describe_server` — confirm the revision and the Henke directory.
2. `list_measurements` → finds `"RuC-07/xrr.dat"` ; `get_measurement` with `max_points: 200` to look at the curve.
3. `assess_xrr` with the nominal structure — is the curve worth fitting: the plateau against the first order, the orders visible, the points per fringe, the zero counts.
4. `calc_reflectivity` with the nominal Ru/C structure (30 periods, $d = 68.5 \text{ \AA}$) at $\lambda = 1.5406 \text{ \AA}$ — compare the Bragg peak position with the measurement.
5. `fit_xrr` with the period of stack 0 and both layer thicknesses free, `smooth: {"passes": 1}` , `theta_range: {"min": 0.3, "max": 4.0}` , `seed: 1` → `job_id: "fit-20260922-143012-a1b2"` .

6. `job_wait` until `finished` (tens of seconds on the CPU, much less on a GPU; the result's `elapsed_s` and `iterations_run` give the real cost).
7. `job_result` → `fitted_structure` (d is `report.period_A` ; Γ follows from the layer thicknesses), `chi2` , and the path of `fit.xrcx` .
8. `save_project` with `name: "RuC-07-fit"` and `curves.job_id: "fit-20260922-143012-a1b2"` → `projects\RuC-07-fit.xrcx` , which you open in X-Ray Calc 3 to inspect and refit.

Practice: The lab's rule of thumb for the LFPSO is 100 iterations with 500 to 1000 particles: population wins over iterations. The server's defaults are 500 particles and 100 iterations, the lower end of that rule. They are not the GUI's defaults (the GUI's own population is 1000); the result's `optimizer_used` lists every value the run used.

Security

- **Sandbox:** the server can only read and write under its working directory. A path outside it is refused, never remapped.
- **Read-only inbox:** no tool has a write path into `inbox\` ; your measured data cannot be altered by an assistant.
- **No network:** the server speaks only over its own stdin and stdout to the client that started it. It opens no ports.
- **Journal:** every call and result is appended to `log\calls.jsonl` ; the file is never truncated by the server.
- **Structured errors:** failures are returned as `{code, message, detail}` . The process never shows a dialog or waits for a key, so it is safe to run unattended.

Smoke Test

The repository ships an end-to-end check, `XRC_MCP\smoke\session.ps1` . It starts the server in a throwaway working directory, calls every one of the 18 tools over one live stdio session (jobs are submitted, polled, waited for with `job_wait` and read; one is cancelled), and checks the Ru/C Bragg-peak angle, seed determinism of the optimizer and the fit, the `path_outside_workdir` refusal, that the inbox is byte-identical afterwards, and that the journal holds exactly one line per call.

```
pwsh -File XRC_MCP\smoke\session.ps1
```

Exit code 0 means every check passed; `-KeepWorkdir` leaves the work directories behind for inspection. It takes a few seconds.

See Also

- [Fitting \(Optimization\)](#) — the LFPSO the server runs
- [Universal Mirror Optimizer](#) — the optimizer behind `optimize_mirror`
- [File Formats](#) — `.xrcx` and data files
- [Source Code & Git](#) — building the server

18 File Formats

Project files (.xrcx), experimental data files, Henke tables, and the settings file

Project File (.xrcx)

A ZIP archive of the project's working folder:

- `project.dsc` — the project tree (binary): every model's structure (stacks, layers, substrate) as JSON, the titles and descriptions of the data items, and the profile extensions with their coefficients.
- `params.dsc` — an INI file with the calculation and fitting settings, no layers: `[PARAMS]` (N, mode, polarisation, chart minimum), `[ANGLE]` and `[WAVE]` (ranges, wavelength, resolution, the 2 θ switch), `[FIT]` and `[LFPSO]` (fit settings, including `SolveScale` and `ScaleWindow`), `[INFO]` Version, and `[STATE]` (linked data, active model, log scale).
- `data_<ID>.dat` — one text file per data item, in the clipboard layout (three header lines, then angle and reflectivity).
- `calc.dat` — the active model's calculated curve.

Data curves are stored with the angle rounded to 0.001° and the reflectivity to five significant digits; Load, Normalize, Smooth and Trim all write through this, so a finer angle step than 0.001° is rounded and can produce repeated angles.

The current project version is 8. A project of version 7 or earlier opens with the substrate density reset to 0 (bulk), because earlier versions ignored the substrate density.

Experimental Data Files

Plain text, one point per line: angle (or wavelength), a single TAB, reflectivity:

```
* Optional description lines starting with *
0.100    1.000000
0.200    0.999850
...
```

Column 1: the angle in degrees, 2 θ when the calculation's 2 θ switch is on and θ when it is off (a text file is not converted), or the wavelength in Å in wavelength mode.

Column 2: reflectivity.

Use exactly one separator (one TAB, or one space in a file without TABs) and no third column; both numbers must start with a digit. Other lines are skipped. The full rules are in [Experimental Data > Data Format](#).

XRDML (PANalytical)

A PANalytical / Malvern `.xrdml` file, schema 1.0 to 2.x, is read without conversion by **Data > Load** and by the MCP server's inbox. What is read:

- the first `scan` of the file that has data points, its `2Theta` positions (start/end, a position list, or `Omega` for a rocking curve; a file whose 2θ does not move and has no Omega positions is refused);
- `counts` (raw, with `beamAttenuationFactors` multiplied in when present) or `intensities` (already corrected), divided by `commonCountingTime` or the per-point `countingTimes` (left undivided when the file has neither);
- the wavelength: $K\alpha_1$ and $K\alpha_2$ with their ratio, weighted together unless a monochromator or a hybrid mirror is in the incident path;
- the detector name and read-out period, the sample id, the scan status and start time.

The curve is normalised to 1 at its maximum. A zero or negative count is replaced by the smallest positive rate before it (1000 counts/s when none precedes it); a text file is floored the same way. The peak count rate of the corrected curve is kept in the description. See [Experimental Data > Loading XRDML Files](#).

Henke Tables

One binary file per material, element or compound, `Henke\<Name>.bin` (the `Henke` folder beside the program, or under the storage folder set in [Settings](#)). A layer's material name must match a file name. Each file holds:

- the material name, the molar mass A (g/mol, per formula unit) and the bulk density (g/cm³);
- records of (photon energy E in eV, f_1 , f_2) on an energy grid. The wavelength is $\lambda = 12398.6 / E$ (Å); f_1 and f_2 are interpolated linearly in E .

The optical constants follow from these: δ and β are proportional to $(\rho / A) \lambda^2 f_1$ and $(\rho / A) \lambda^2 f_2$, with ρ the layer's density, or the bulk density from the file when the layer's is 0. For a compound, f_1 , f_2 and A are all per formula unit, so they must be changed together (see [Edit Henke Table](#)). The tables are included with the application.

Settings File (xrc3.ini)

A standard Windows INI file, `%APPDATA%\X-RayCalc3\xrc3.ini`, or next to the program when a file named `uselocaldata` sits there. Sections:

- `[Path]` — the Henke, storage, benchmark, project, output and jobs folders
- `[Window]` — window state, position and size
- `[Options]` — `CheckForUpdates`, `UpdateInfoURL`, `AutoCalc`, `AutoSave`, `LiveUpdate`
- `[Graphics]` — `LineWidth`
- `[Calc]` — `NumberOfThreads` (0 = all), `BenchmarkRuns` (20), `UseGPU` (on). The calculation parameters themselves (range, wavelength, resolution, fit settings) are stored in each project's `params.dsc`, not here.
- `[Recent]` — the recent-projects list

See Also

- [Projects](#)
- [Experimental Data](#)
- [Application Settings](#)

19 Illustrative Examples

Single-layer coatings, periodic mirrors, polynomial fitting, and a supermirror

The following examples from the literature demonstrate the application of XRC3 to different types of structures (Penkov *et al.*, 2024).

Single-Layer Coating

Evaluation of a single-layer TiZrNi coating deposited by DC magnetron sputtering. The XRR was fitted starting with a two-layer model (TiZrNi + TiO₂), which yielded a cost function of ~0.48. Adding extra layers minimized the cost function to 0.14, revealing non-stoichiometric surface layers and hydrocarbon contamination.

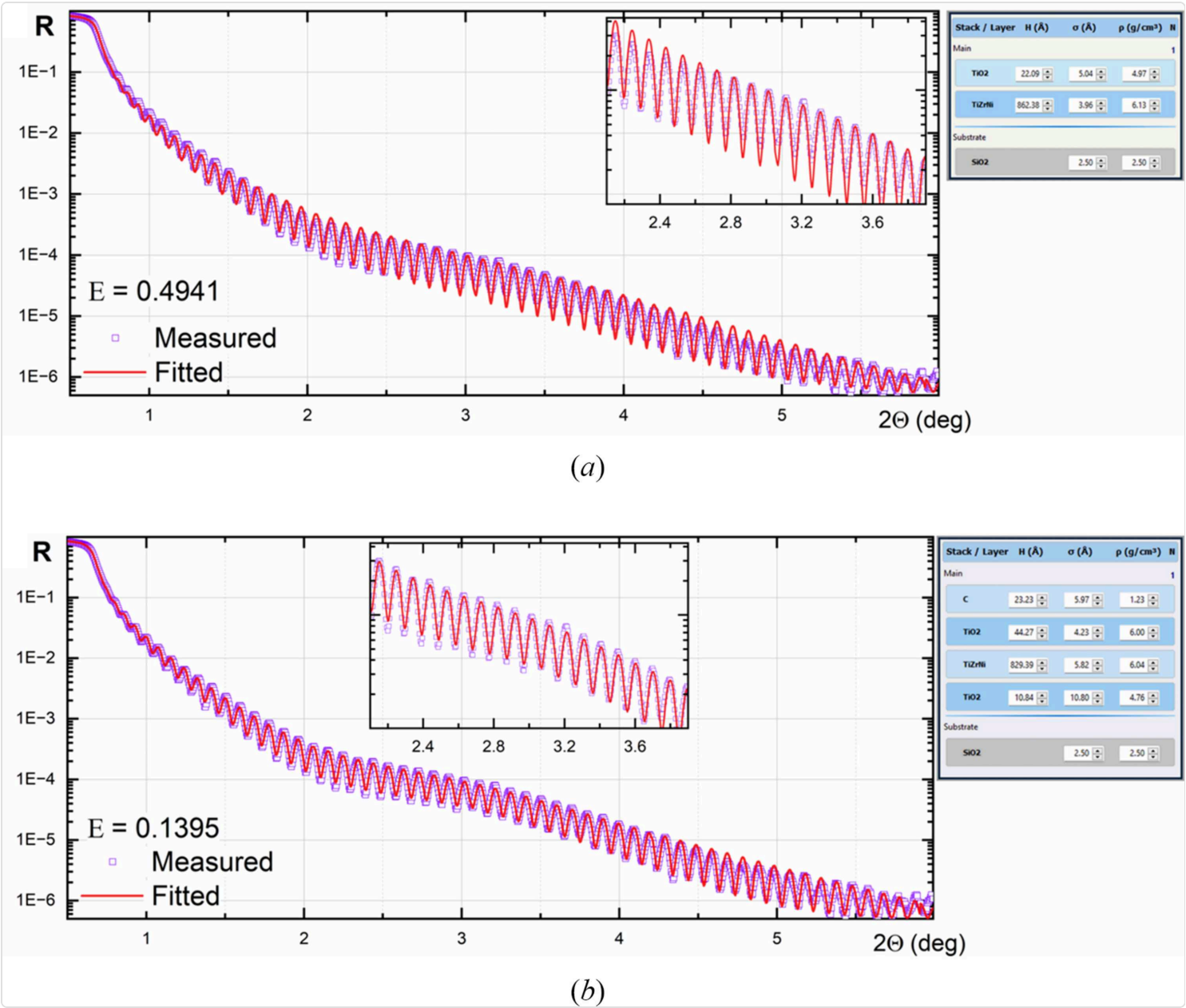


Figure 5. XRR curve fitting for a single-layer coating. (a) Two-layer model. (b) Four-layer model. Insets show the fitted structures. (From Penkov *et al.*, 2024.)

Periodic X-Ray Mirrors

Fitting of XRR curves for Mo/Si and Mo/B periodic multilayer mirrors (PMMs) in periodic mode. The Mo/Si results reveal the well-known asymmetric intermixed zones between Mo and Si. For the 2.3 nm period Mo/B PMM, fitting revealed thin asymmetric intermixing zones and a density difference between the thick Mo sublayer (near-bulk density) and the 0.6 nm Mo layers within the periodic structure (84% of bulk).

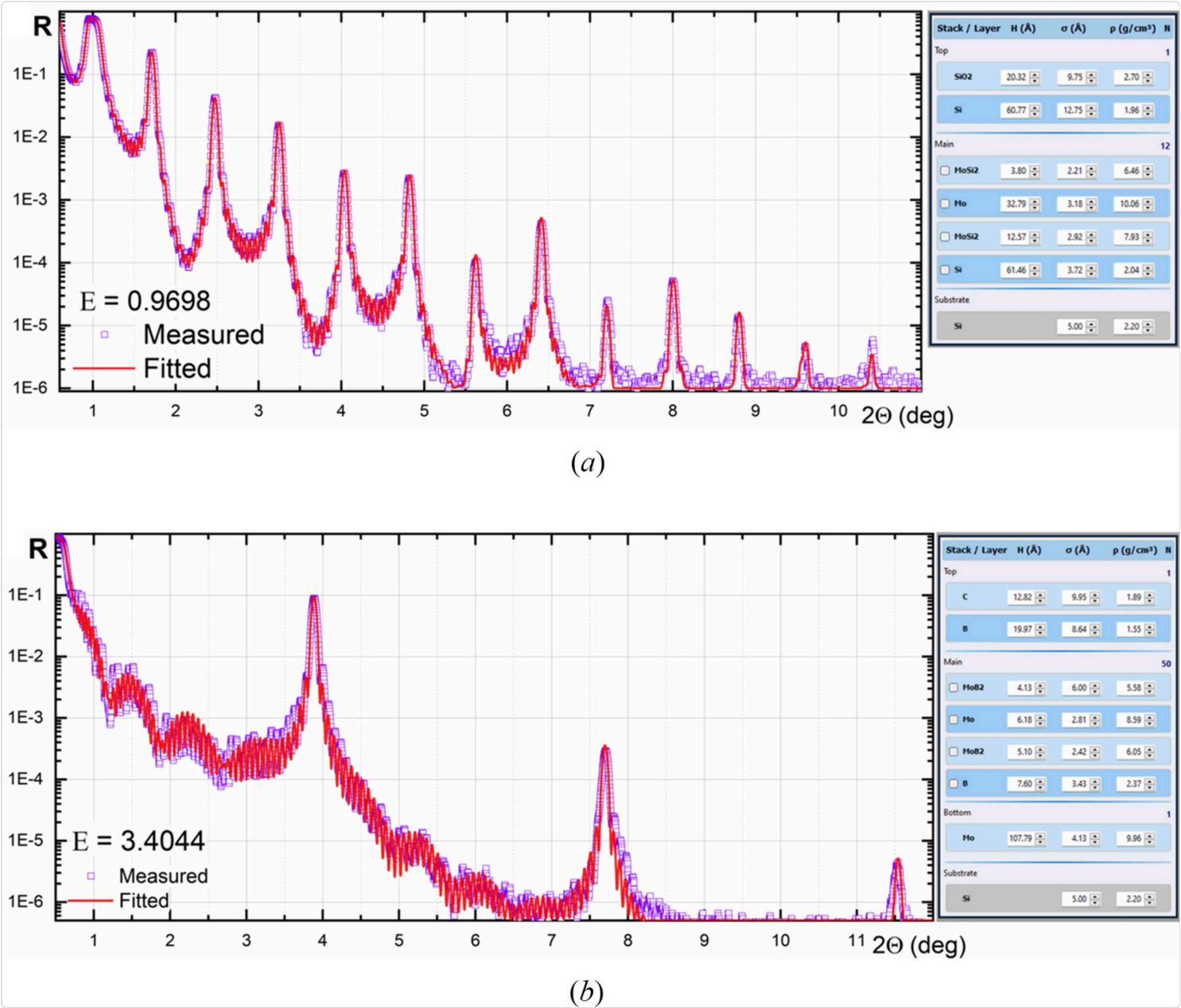


Figure 6. XRR curve fitting for X-ray mirrors. (a) Mo/Si. (b) Mo/B. (From Penkov et al., 2024.)

Polynomial Fitting on Graded PMMs

A W/BN PMM fitted in polynomial mode. The broadening and shape changes of diffraction peaks indicate variation of layer parameters across the stack. Due to unstable deposition conditions, the thickness and roughness of layers were found to vary gradually.

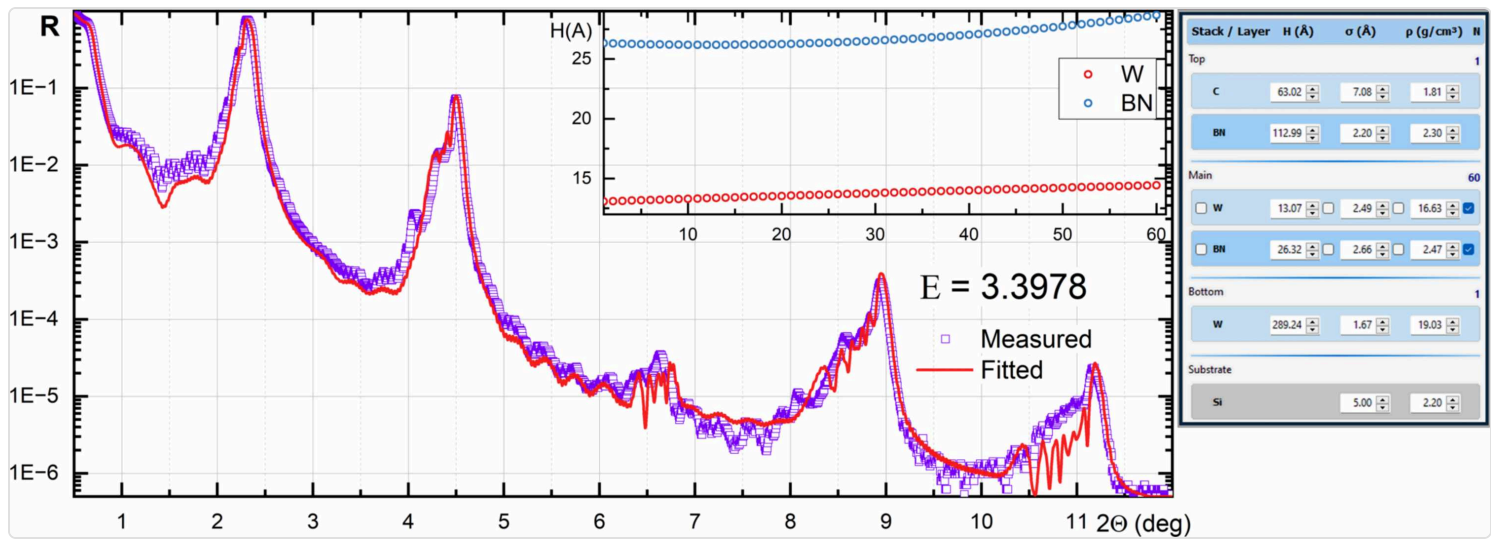


Figure 7. Polynomial XRR curve fitting for a W/BN X-ray mirror with gradually changed thickness of layers. (From Penkov et al., 2024.)

Supermirror

An Sb/B₄C supermirror fitted in irregular mode with paired roughness and density. The fitting revealed the designed depth-graded structure.

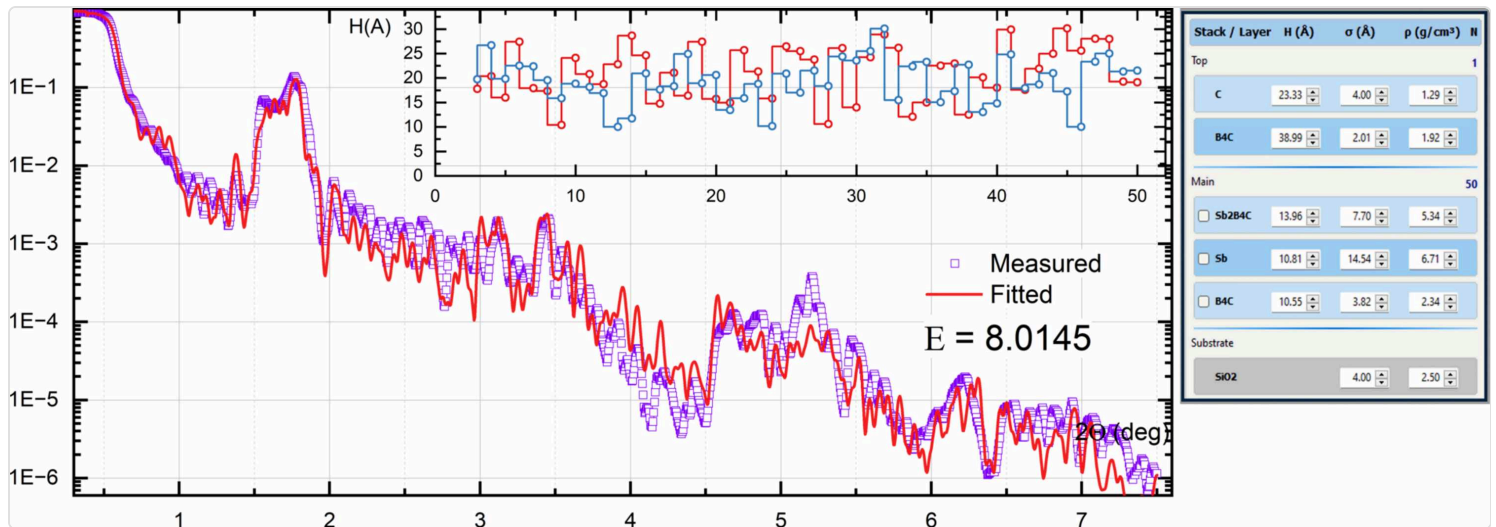


Figure 8. XRR curve fitting for an Sb/B₄C supermirror. (From Penkov et al., 2024.)

See Also

- [Fitting \(Optimization\)](#)
- [Quick Start](#)

20 Troubleshooting

Crashes, x32 versus x64, locked projects, convergence, slow calculation, data import

Application crashes without error message

This is typically caused by running out of memory when solving very large fitting problems (many layers or a very large population size).

Solution: Use the **x64 version** (`XRayCalc3.x64.exe`). It is somewhat slower than the x32 version for small models, but can address much larger amounts of RAM. The x32 version is limited to approximately 2 GB of memory.

Choosing between x32 and x64

Both versions are included in the distribution:

VERSION	ADVANTAGE	LIMITATION
x32 (<code>XRayCalc3.exe</code>)	~2× faster calculation	Limited to ~2 GB RAM
x64 (<code>XRayCalc3.x64.exe</code>)	Handles large models and populations	Slower for small models

Use **x32** for routine calculations with small-to-medium models. Switch to **x64** when working with large multilayer structures, high population sizes, or when you encounter crashes.

The application shows a “project locked” message

Another instance of X-Ray Calc 3 may have the project open, or a previous session crashed without releasing the lock. Close the other instance, or manually delete the `temp.lock` file from the temp directory (`%TEMP%\X-RayCalc3\`).

Fitting does not converge

- **Widen the fit limits** — the initial parameter bounds may be too narrow to contain the true solution.
- **Increase population size** — more particles explore the parameter space more thoroughly.
- **Increase iterations** — the algorithm may need more time to converge.
- **Enable Shake** — this helps escape local minima.
- **Check initial model** — a better starting guess leads to faster convergence.
- **Try a different fitting mode** — Periodic mode may work better than Irregular for regular multilayers, and vice versa.
- **Check the Bragg peak positions first** — a Periodic fit holds the period D of every repeating stack at its start value, so it cannot move the peaks. Set the period so the peaks line up, or fit in Irregular or Polynomial mode (see [Fitting Modes](#)).

- **Look at the scale pane** — if it reads *at bound*, the normalisation is off by more than the Window allows: renormalise, or widen the Window (see [Solving the Measured Scale](#)).
 - **Check the chart's lower reflectivity limit** — the calculated curve is floored at it, so measured points below it are compared with that limit.
 - **Link the data** — the fit needs the measured curve linked to the model and visible; otherwise it stops with *Measured curve is not linked!*
-

Calculation is slow

- Ensure multi-threading is enabled (**Settings > Calc & Fit > Auto (Use all)**).
 - Reduce the number of calculation points (N) for preliminary calculations. N is used only when no measured curve is linked (or it is hidden); with linked data the model is calculated at the measured angles.
 - For fitting, turn on **Settings > Calc & Fit > Use the GPU for fitting** (see [GPU Acceleration](#)).
 - Use the **x32 version** for small-to-medium models — it is approximately 2× faster than x64.
 - Use the **x64 version** only when the model is too large for x32 (memory limit).
-

Material not found

- Check the spelling (chemical formula must be exact, e.g., "SiO2" not "Sio2").
 - The material may not exist in the Henke database. Use **Tools > Create New Material** to add it.
-

Chart appears empty after calculation

- Check that the angle/wavelength range is appropriate for the structure.
 - For very thin films, use small angles (0.01 to 5 degrees).
 - Switch between linear and logarithmic scale — the reflectivity may be visible only on one scale.
 - Verify that the model has at least one layer defined.
-

Data import fails

- Ensure the data file is plain text with two numeric columns.
- Remove any header lines that are not prefixed with `#`.
- Check that decimal separators match your system locale (period vs. comma).
- Try pasting the data via clipboard instead.

See Also

- [Installation & First Launch](#)
- [Fitting \(Optimization\)](#)
- [Online Wiki](#)

21 Source Code & Git

The GitHub repository: getting the source, branches, building, dependencies, tests, and contributing

The Repository

X-Ray Calc 3 is open source. The complete source code, the issue tracker, the release installers and the [wiki](#) live in one Git repository on GitHub:

```
https://github.com/OleksiyPenkov/X-RayCalc3
```

The repository holds more than the GUI. It is a Delphi project group with four programs that share one calculation engine:

PROGRAM	PURPOSE
XRayCalc3	The GUI application this manual describes (<code>XRayCalc3.exe</code> , <code>XRayCalc3.x64.exe</code>).
xrccmd	Command-line calculation, fitting, and the Universal Mirror Optimizer .
XRFCalc	GUI for viewing and running universal-mirror optimizations (<code>.xrfx</code> result files).
XRC_MCP	The MCP server that exposes the engine to AI assistants.

Releases: You do not need the source to use the program. Windows installers are published under **Releases** on the repository page; each installer ships the 32-bit and 64-bit GUI, the Henke optical-constant tables and the example projects.

Getting the Source

Install **Git for Windows**, then clone the repository:

```
git clone https://github.com/OleksiyPenkov/X-RayCalc3.git
cd X-RayCalc3
```

To update a clone you already have:

```
git pull
```

If you only want to read the code, the **Code > Download ZIP** button on GitHub gives you a snapshot of the current branch without Git.

Branches and History

REF	CONTENT
master	The development branch since 2023. Releases are built from it. This is what <code>git clone</code> checks out.
main	The earlier public history with the flat directory layout and the 3.5.x releases, kept for reference.
tag legacy-main-3.5.3	The last state of that earlier history.

Commit messages follow a small convention: a leading `+` marks a new feature, a leading `*` a modification or fix. `git log --oneline` is therefore a readable change list.

Repository Layout

XRayCalc3/	GUI application
Forms/	Main windows (frm_Main, settings, about, ...)
Views/	Reusable frames (chart info, project panel, ...)
Units/	Business logic (config, types, helpers, orchestrators)
LFPSO/	Particle swarm optimization for curve fitting
Components/	Custom VCL components + package (tree, grid, editors)
Editors/	Data editor dialogs (profile, Henke table, JSON, ...)
Assets/	Icons, this help, development plans
Tests/	DUnitX test suite
XRC_CMD/	xrccmd, the command-line interface
XRFCalc/	XRF mirror-design GUI
XRC_MCP/	MCP server for AI assistants
Shared/	
Math/	Calculation engine, complex math, materials database
Universal/	Universal-mirror types, fitness, templates, XRF lines
_Out/	Build output (BIN/, DCU/, DCU64/)
_Installer/	Inno Setup script and deployment script
docs/	Design documents and implementation plans
XRC3.groupproj	Delphi project group

Project files are `.xrcx` archives (see [File Formats](#)). Lengths are in ångström, densities in g/cm³, angles are the grazing angle θ in degrees, throughout the code.

Building

The programs are written in Delphi (Object Pascal) and build with **Embarcadero RAD Studio 37.0** (Delphi 13). Win32 is the primary target for the GUI and the CLI; XRC_MCP is built for Win64 only.

In the IDE

1. Open `XRC3.groupproj` in RAD Studio.

2. Choose the **Release** configuration and the **Win32** platform (or Win64 for the 64-bit GUI).
3. Right-click the project group in the Project Manager and select **Build All**. The build order is XRayCalc3 → XRayCalcVisualControls → xrccmd → XRFCalc → XRC_MCP.

From the command line

Set the Embarcadero environment variables, then call MSBuild for the project you need. Always use `/t:Build` ; `/t:Make` does not work with these projects.

```
set BDS=C:\Program Files (x86)\Embarcadero\Studio\37.0
set BDSCOMMONDIR=C:\Users\Public\Documents\Embarcadero\Studio\37.0
set MSB=C:\Windows\Microsoft.NET\Framework\v4.0.30319\msbuild.exe

%MSB% XRayCalc3\XRayCalc3.dproj /t:Build /p:Config=Release /p:Platform=Win32
%MSB% XRC_CMD\xrccmd.dproj      /t:Build /p:Config=Release /p:Platform=Win32
%MSB% XRFCalc\XRFCalc.dproj     /t:Build /p:Config=Release /p:Platform=Win32
%MSB% XRC_MCP\XRC_MCP.dproj     /t:Build /p:Config=Release /p:Platform=Win64
```

Executables land in `_Out\BIN\` . The 64-bit GUI is `XRayCalc3.x64.exe` ; `XRayCalc3.exe` is the 32-bit build.

Git on PATH: XRC_MCP regenerates `XRC_MCP\units\gitrev.inc` from `git describe --always --dirty` in its pre-build event, so `git` must be on `PATH` when you build it. The value is what the server reports as its revision; a `-dirty` suffix means the tree had uncommitted changes when it was built.

Dependencies

The following libraries must be on the Delphi library path for the platform you build:

LIBRARY	USED FOR
OmniThreadLibrary 3.08 or later	Parallel calculation and fitting. Older versions truncate 64-bit code pointers in <code>OtlTaskControl.pas</code> , and every Win64 build that runs the optimizer or the fit then hangs at the first iteration. github.com/gabr42/OmniThreadLibrary
FastMath	Vectorised math. github.com/neslib/FastMath
Raize Components (Konopka Signature VCL Controls)	UI controls (<code>Rz*</code> prefix).
Virtual TreeView	Project tree and grids.
Abbrevia	The <code>.xrcx</code> ZIP archives.
SynEdit	The JSON and text editors.
DUnitX	The test suite; shipped with RAD Studio.

Symptom of an old OmniThreadLibrary: a Win64 fit or optimization that stops at iteration 0 and a process that will not exit. Win32 builds are unaffected. Check out the library at tag `release-3.08` or later and rebuild.

Running the Tests

`XRayCalc3\Tests\XRayCalc3Tests.dproj` is a DUnitX suite (Win32, Debug) covering the calculation engine, the Henke tables, the LPSO fit, the universal optimizer and every XRC_MCP unit. Build it, then run:

```
%MSB% XRayCalc3\Tests\XRayCalc3Tests.dproj /t:Build /p:Config=Debug
XRayCalc3\Tests\_Out\BIN\XRayCalc3Tests.exe --exitbehavior:Continue
```

The XRC_MCP server additionally has an end-to-end smoke test, described in [MCP Server > Smoke Test](#).

Contributing

- **Bug reports and feature requests** go to the **Issues** tab of the repository. Attach the project file (`.xrcx`) and the measured data that reproduce the problem, and state whether you ran the 32-bit or the 64-bit build (see [Troubleshooting](#)).
- **Code changes** are welcome as pull requests against `master` . Fork the repository on GitHub, create a branch, commit with the `+ / *` prefix convention, and open the pull request. Build the Win32 Release target and run the test suite before submitting.
- **Documentation** changes to this manual are code changes too: the pages live in `XRayCalc3/Assets/Docs/Help/` . Tutorial-style material belongs in the [wiki](#), which anyone with a GitHub account can edit.

Licence and Citation

X-Ray Calc 3 is distributed under the **GNU General Public License v3.0**; see the `LICENSE` file in the repository.

If you use the software in published work, please cite:

- Penkov, O. V., Li, M., Mikki, S., Devizenko, A. & Kopylets, I. (2024). *X-Ray Calc 3: improved software for simulation and inverse problem solving for X-ray reflectivity*. J. Appl. Cryst. **57**, 555–566. doi:10.1107/S1600576724001031
- Penkov, O. V., Kopylets, I. A., Khadem, M. & Qin, T. (2020). *X-Ray Calc: a software for the simulation of X-ray reflectivity*. SoftwareX **12**, 100528. doi:10.1016/j.softx.2020.100528

See Also

- [Online Wiki](#) — tutorials and notes hosted next to the repository
- [MCP Server](#) — the fourth program in the project group
- [Introduction](#) — what the software does

22 Online Wiki

The GitHub wiki that accompanies this manual: its pages, and how to read and edit them

Overview

Besides this manual, which is installed with the program and describes every feature, X-Ray Calc 3 has an **online wiki** hosted with the source repository on GitHub:

```
https://github.com/OleksiyPenkov/X-RayCalc3/wiki
```

The wiki is the place for material that changes faster than a release, or that is easier to write as a walkthrough than as a reference: tutorials with screenshots, worked fitting sessions, benchmark results, and answers to questions that come up repeatedly. It is public, needs no login to read, and can be edited by anyone with a GitHub account.

Which one to read: Start with the wiki's **Getting started** if you are new to the program and want to see a complete session end to end. Use this manual when you need to know what a particular setting, button or file format does.

Wiki Pages

PAGE	CONTENT
Home	Entry page with links to every other page.
Getting started	A first session: open a demonstration project, run a calculation from the Computation tab, edit layers, stacks and model properties, change polarization and the angle range, and export or plot the result.
How to create a model and calculate XRR	Building a multilayer model from an empty project: stacks, layers, materials from the Henke database, and the calculation settings.
Working with experimental data	Loading a measured curve, the expected file format, and comparing it with the calculated one.
Working with experimental data (Data conditioning)	Preparing a measured curve for fitting: normalize, smooth and trim, in that order, with the reasoning behind each step.
Manual curve fitting	Fitting by hand before (or instead of) the automatic fit: create an initial model and load the data, roughly match the period of the structure, then adjust the roughness of the layers until the peak intensities agree.
Benchmarks	Convergence studies of the fitting algorithm: the effect of the algorithm constants, LFPSO versus Shake LFPSO on several structures, and how the iteration count and population size influence convergence, with CPU timings. A benchmark archive can be downloaded from the repository.
Troubleshooting	Known problems and their remedies, for example the application closing without an error message when a large fit runs out of memory in the 32-bit build.
User Manual	An older, interface-oriented description of the main menu and toolbars. Superseded by this manual, kept for readers of earlier versions.

This Manual and the Wiki

Several wiki pages have a counterpart here. The wiki version is the walkthrough; the chapter here is the reference.

WIKI PAGE	CHAPTER IN THIS MANUAL
Getting started	Quick Start
How to create a model and calculate XRR	Defining a Model, Calculation
Working with experimental data	Experimental Data, File Formats
Working with experimental data (Data conditioning)	Experimental Data > Data Conditioning Walkthrough
Manual curve fitting	No counterpart: the technique is described only in the wiki. Fitting covers the automatic fit.
Benchmarks	No counterpart. Tools > Benchmark describes how to run the built-in benchmark on your own machine.
Troubleshooting	Troubleshooting

Tip: The [Help > User Manual](#) menu command opens this manual in your browser. The wiki is at the address at the top of this page; the repository's README links to it as well.

Editing the Wiki

Corrections and new tutorials are welcome. GitHub wikis are edited in the browser:

1. Sign in to GitHub (a free account is enough).
2. Open the page you want to change and click **Edit** in the top right, or click **New page** on any wiki page to add one.
3. Write in Markdown: `#` headings, `*` bullets, `![caption](image-url)` for screenshots. The **Preview** tab shows the rendered result.
4. Enter a short edit summary and click **Save page**. The change is live at once.

Screenshots are added by dragging the image into the editor, which uploads it and inserts the Markdown link for you. Please keep to the existing page structure: one topic per page, a short introduction, then numbered steps.

What belongs where: Walkthroughs, tips, benchmark results and questions-and-answers go in the wiki. A description of what a setting does, a file format, or a keyboard shortcut belongs in this manual, which is versioned with the source code (see [Source Code & Git > Contributing](#)).

Cloning the Wiki

A GitHub wiki is itself a Git repository, so you can keep an offline copy, edit pages in your own editor, or back it up:

```
git clone https://github.com/OleksiyPenkov/X-RayCalc3.wiki.git
```

Each page is one Markdown file named after its title (`Getting-started.md` , `Manual-curve-fitting.md` , and so on). Commit and push to publish your edits; pushing requires write access to the wiki, which every signed-in GitHub user has unless the repository owner restricts it.

See Also

- [Source Code & Git](#) — the repository the wiki belongs to
- [Quick Start](#) — the manual's own tutorial
- [Troubleshooting](#)